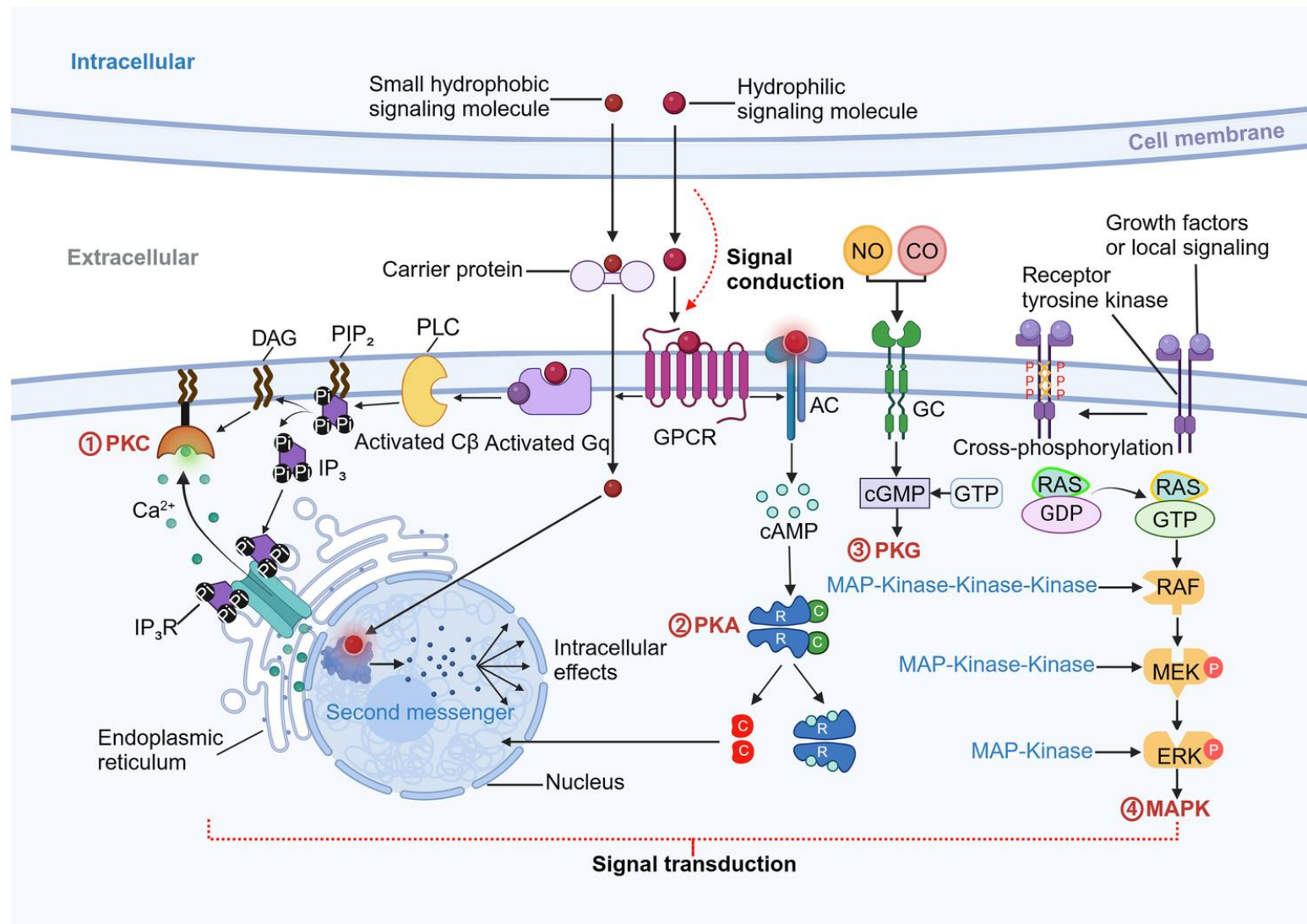


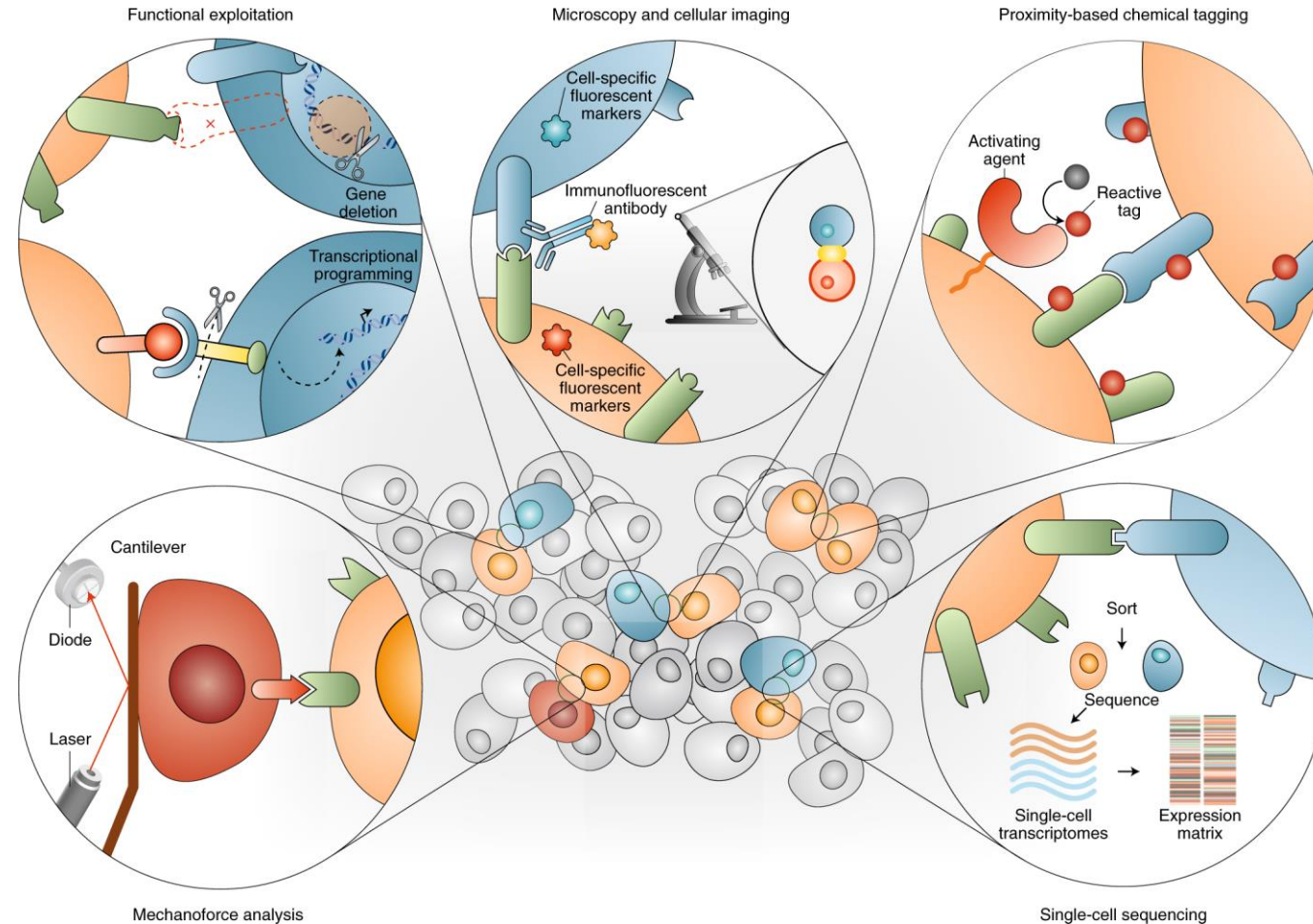
Cell-cell communication lecture

MGC – BioSB Spatial Course 2025

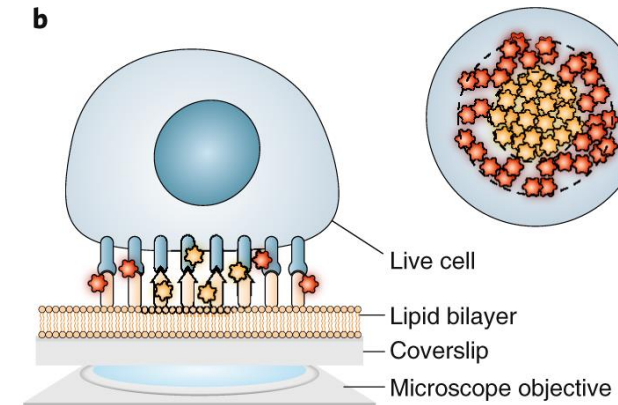
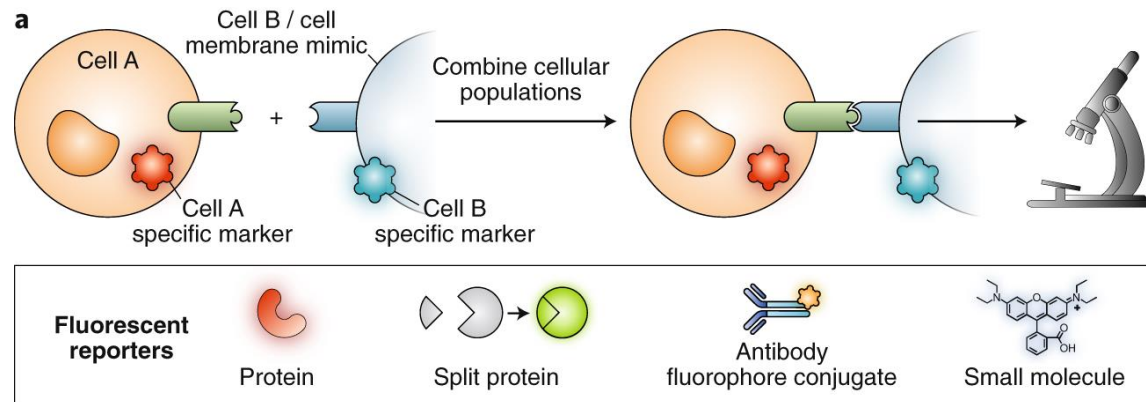
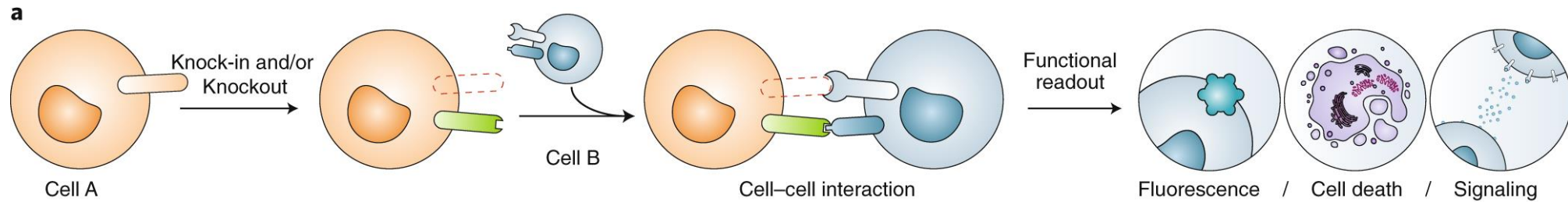


Cellular communication is the process of signal construction to signal transduction.

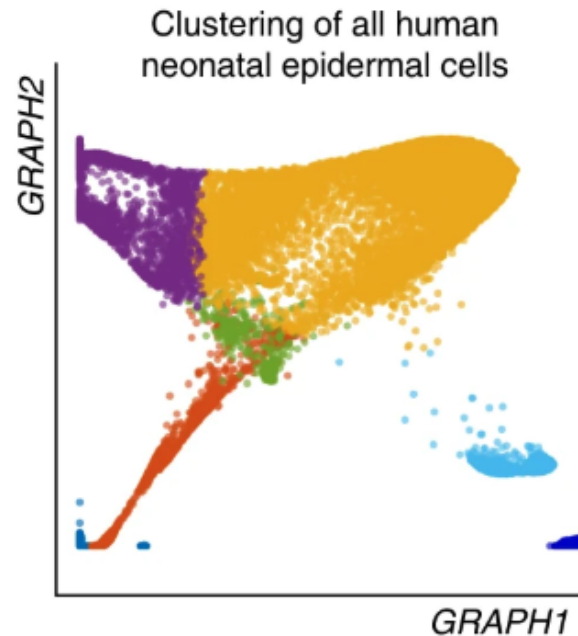
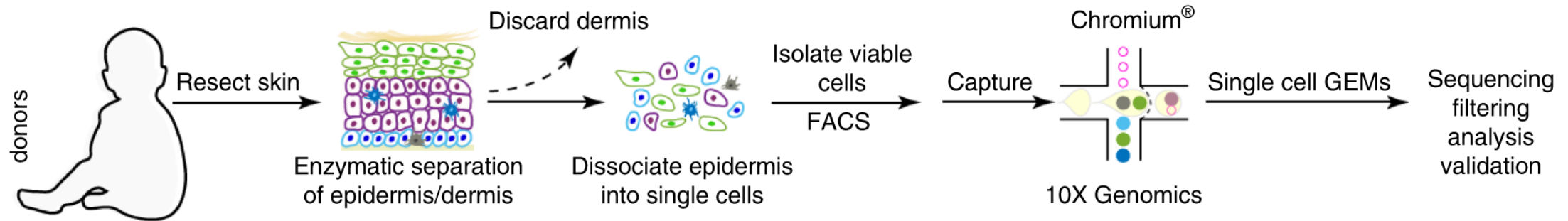
How to study cell-cell communication?



How to study cell-cell communication?



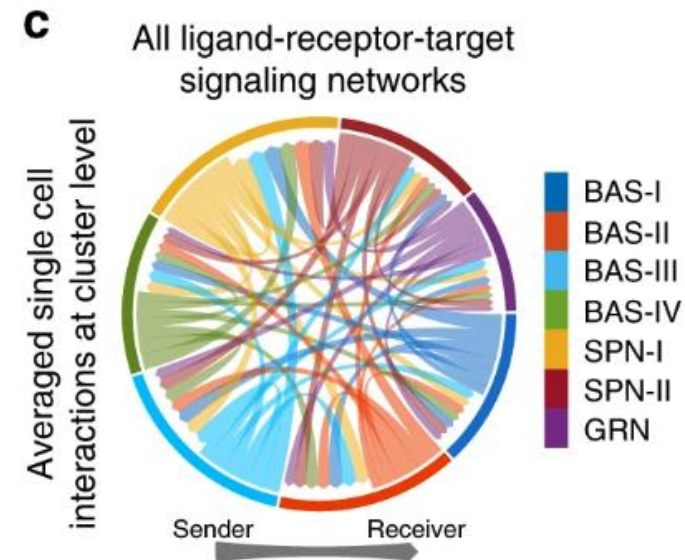
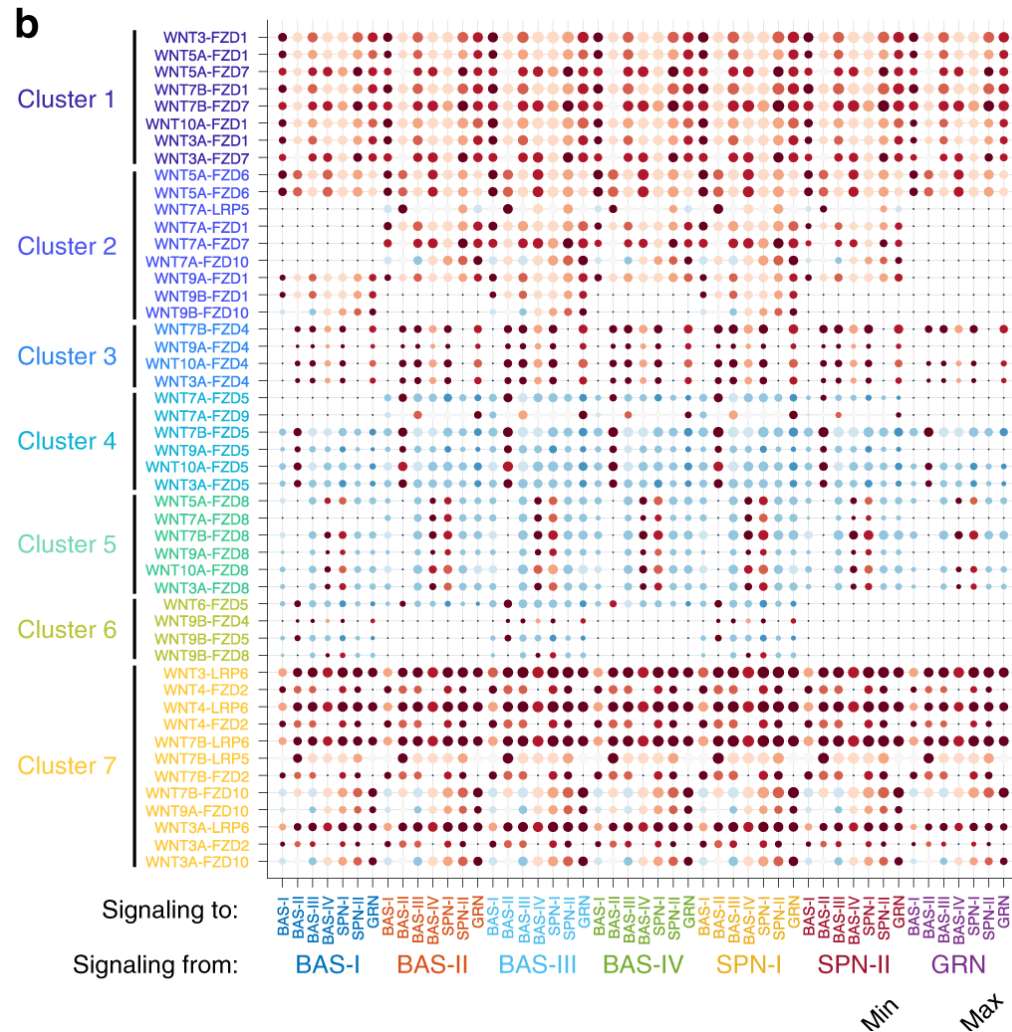
Why study cell-cell communication spatially?



- Putative cellular communities
- Basal Type I - (BAS-I) ~4%
 - Basal Type II - (BAS-II) ~9%
 - Basal Type III - (BAS-III) ~7%
 - Basal Type IV - (BAS-IV) ~3%
 - Spinous - (SPN) ~54%
 - Granular - (GRN) ~16%
 - Melanocytes - (MEL) ~6%

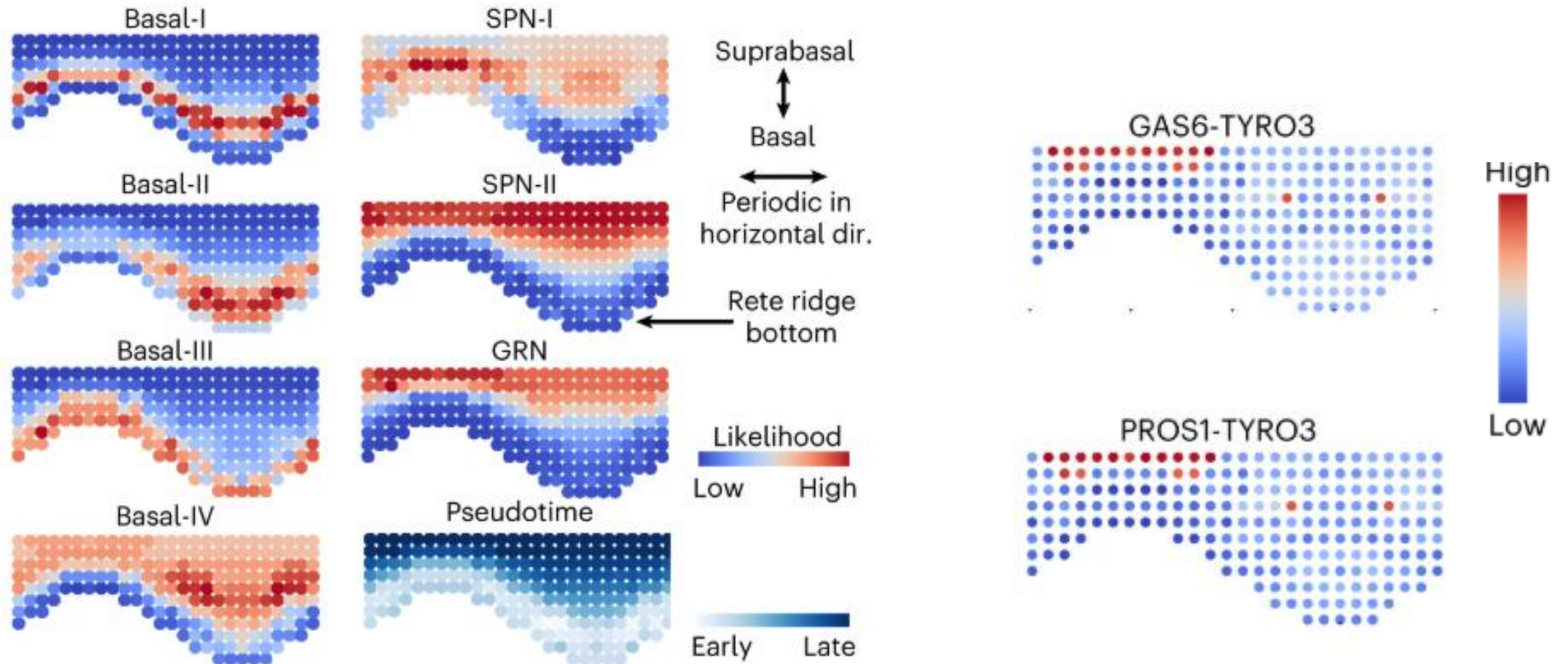
Cell cohort proportions (%)	
BAS (~24%)	GRN (~16%)
SPN (~54%)	MEL (~6%)

Why study cell-cell communication spatially?

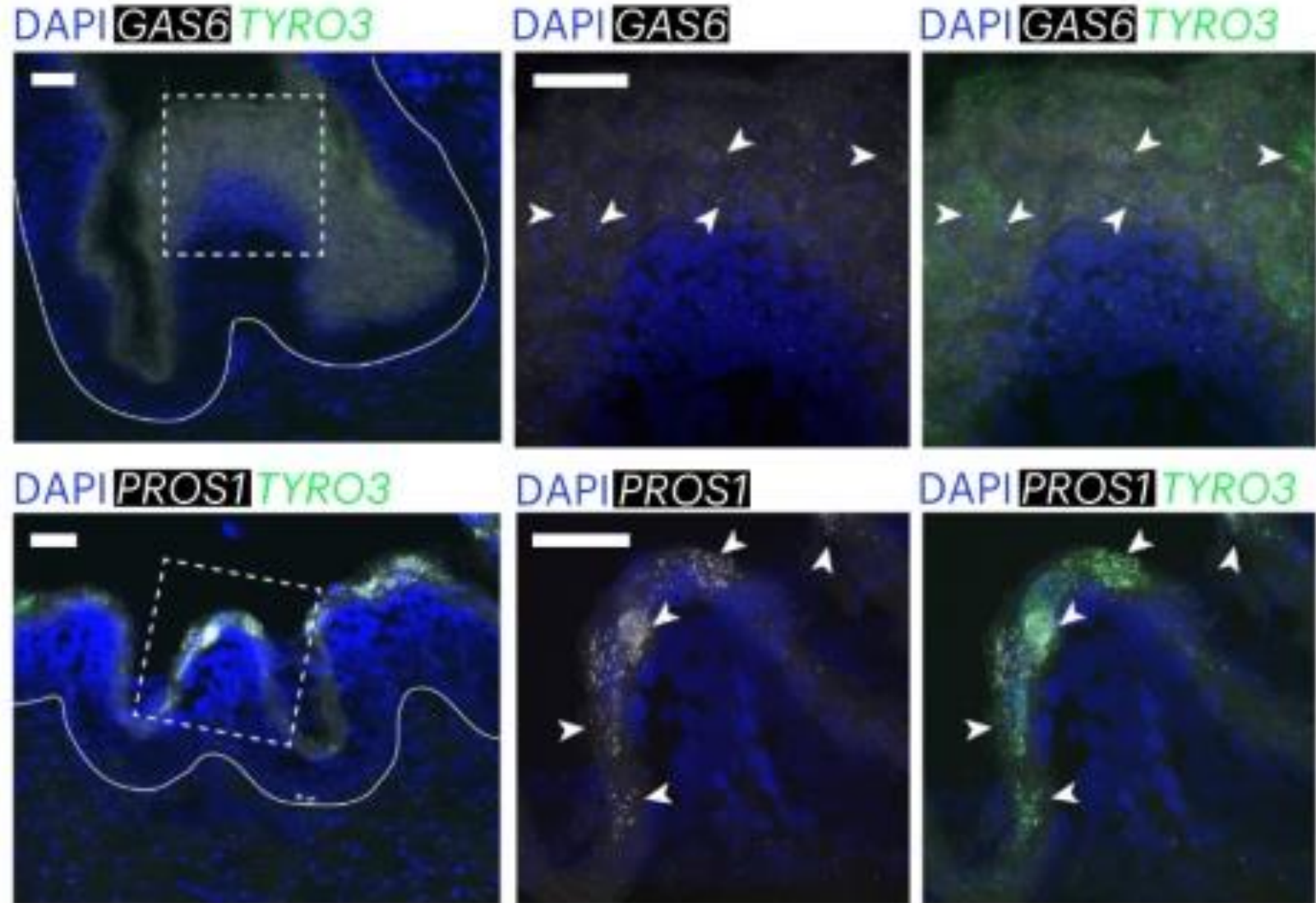
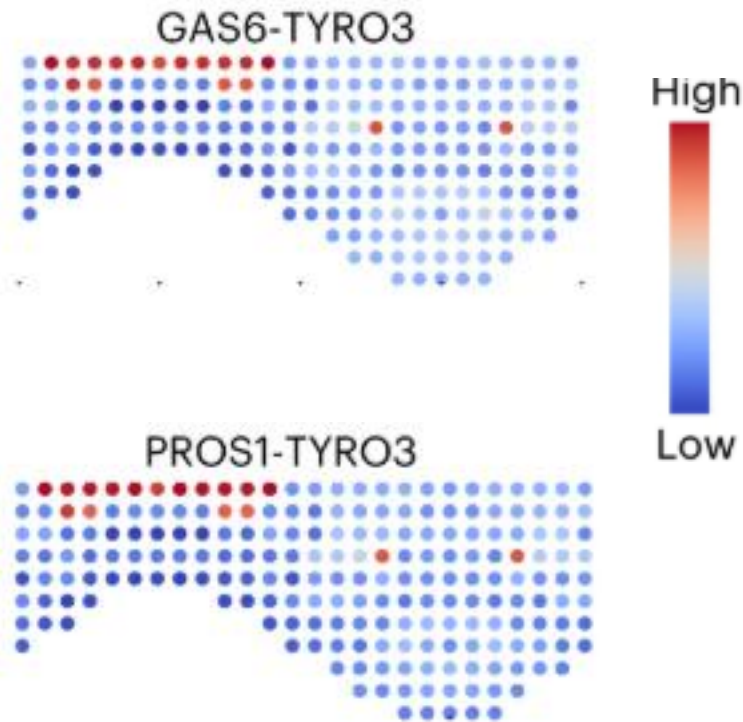


At the single-cell level, WNT signalling appears to be active in **most of the basal and spinous populations**

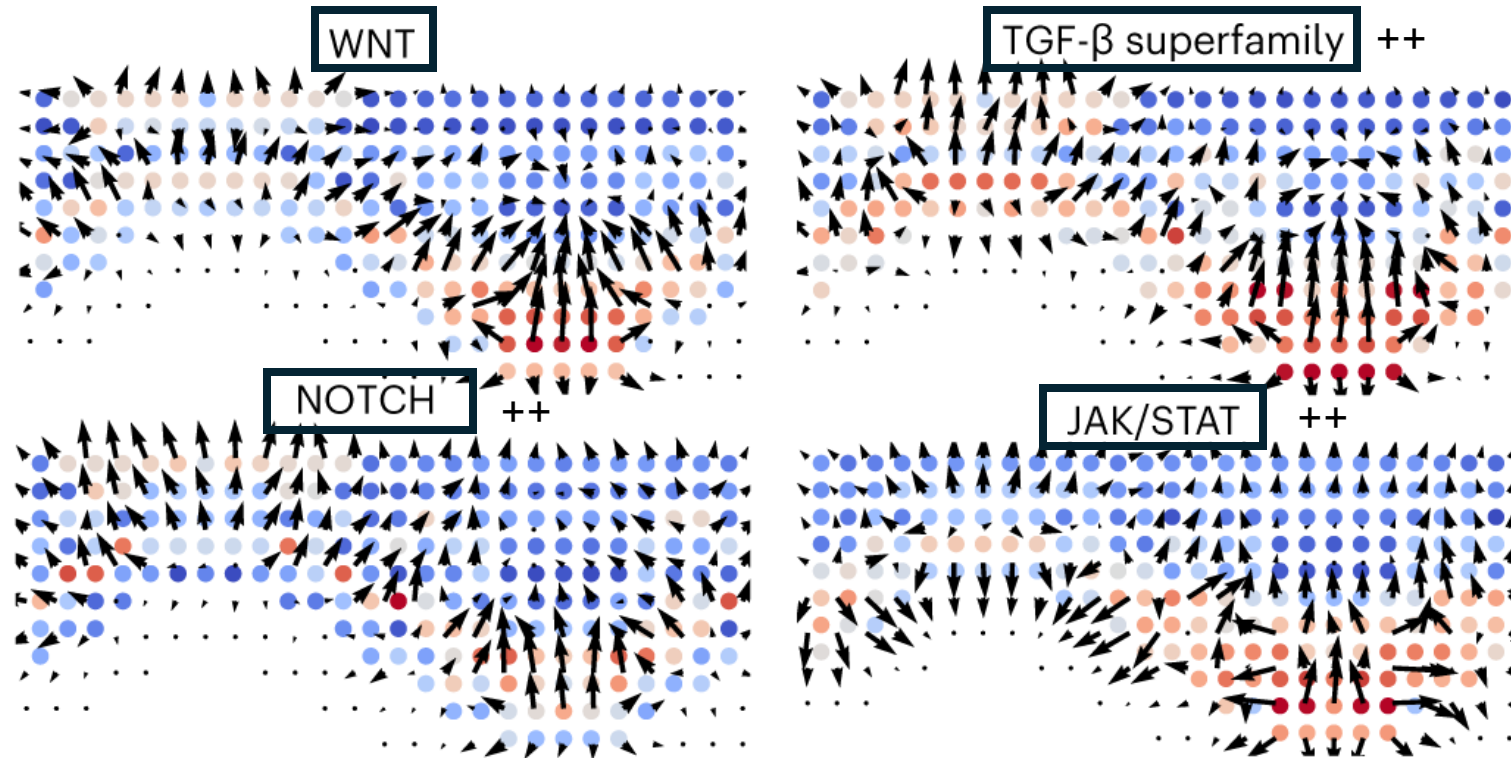
Why study cell-cell communication spatially?

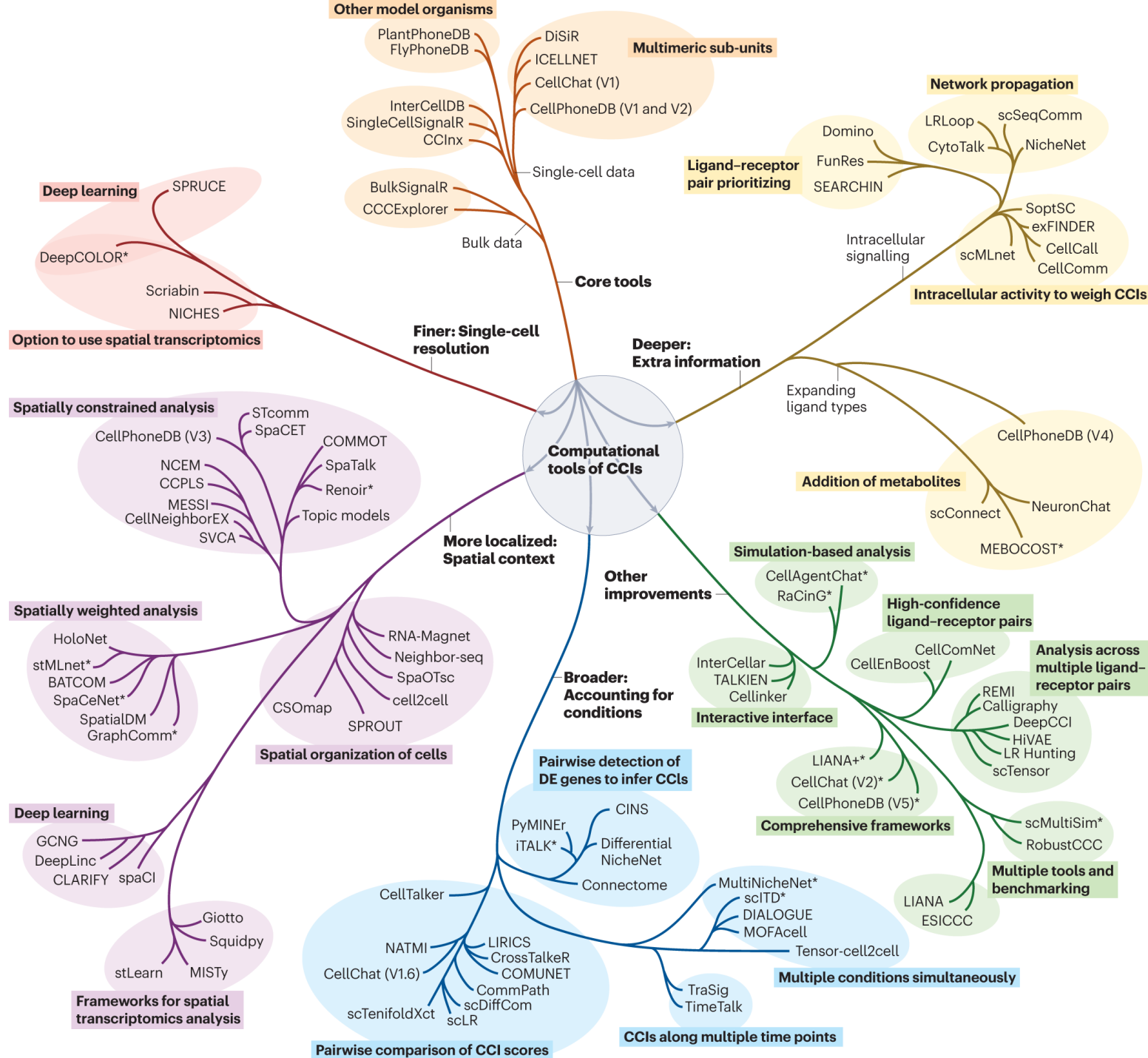


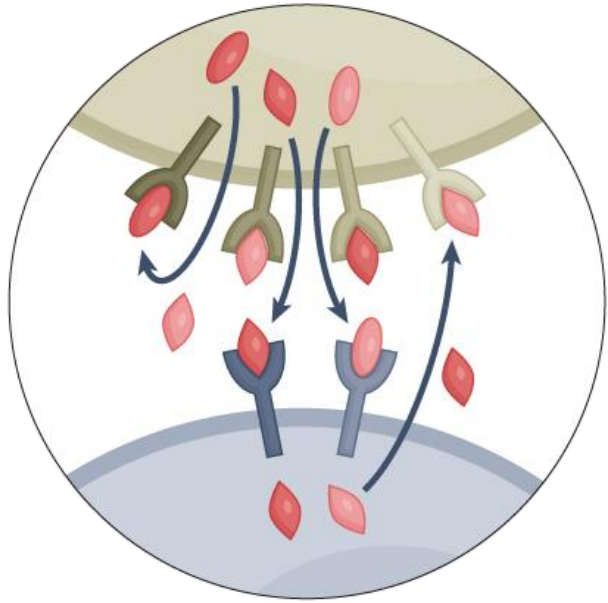
Why study cell-cell communication spatially?



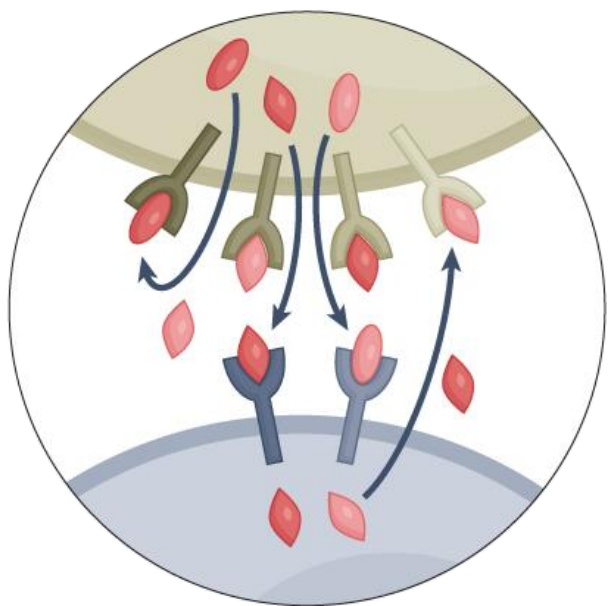
Why study cell-cell communication spatially?



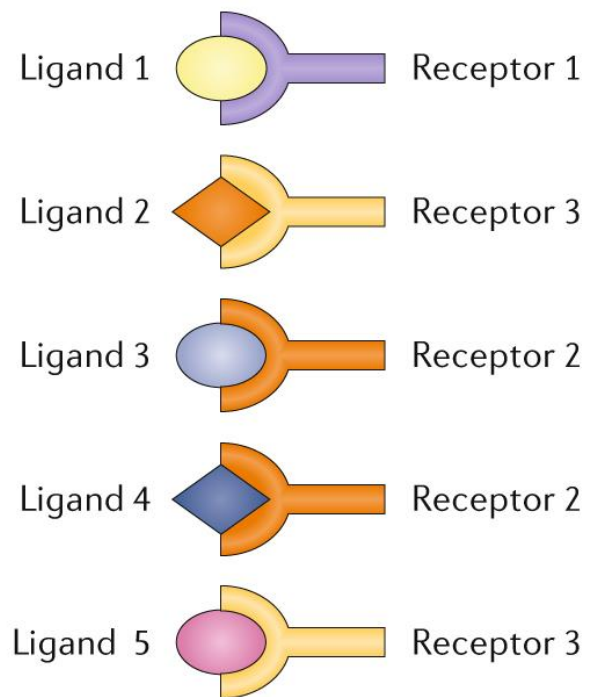




Cell-cell communication from a computational perspective



Prior knowledge



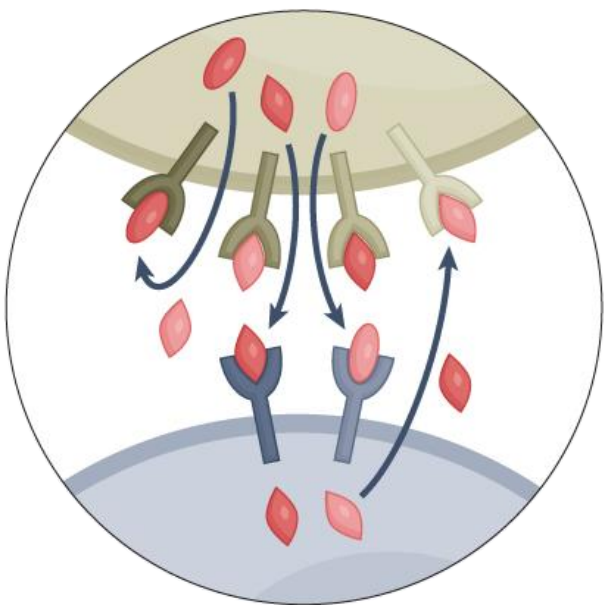
Gene expression

	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6
Ligand 1	3.81	3.46	4.32	2.32	3.00	2.58
Ligand 2	3.17	1.58	2.32	2.32	0.00	3.46
Ligand 3	3.00	3.46	4.91	2.32	6.64	1.58
Ligand 4	4.00	5.32	5.64	7.71	8.02	7.91
Ligand 5	5.32	3.32	4.32	3.91	2.32	5.49
Receptor 1	4.70	4.58	4.17	3.46	3.81	3.58
Receptor 2	3.00	4.09	4.46	6.91	7.13	6.78
Receptor 3	4.32	2.00	3.00	3.17	1.00	3.81

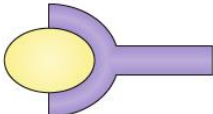
Expression value

8

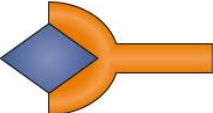
0

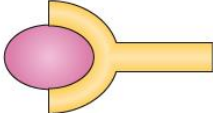


Prior knowledge

Ligand 1  Receptor 1

Ligand 2  Receptor 3

Ligand 4  Receptor 2

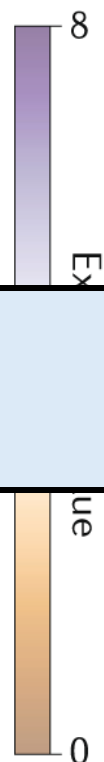
Ligand 5  Receptor 3

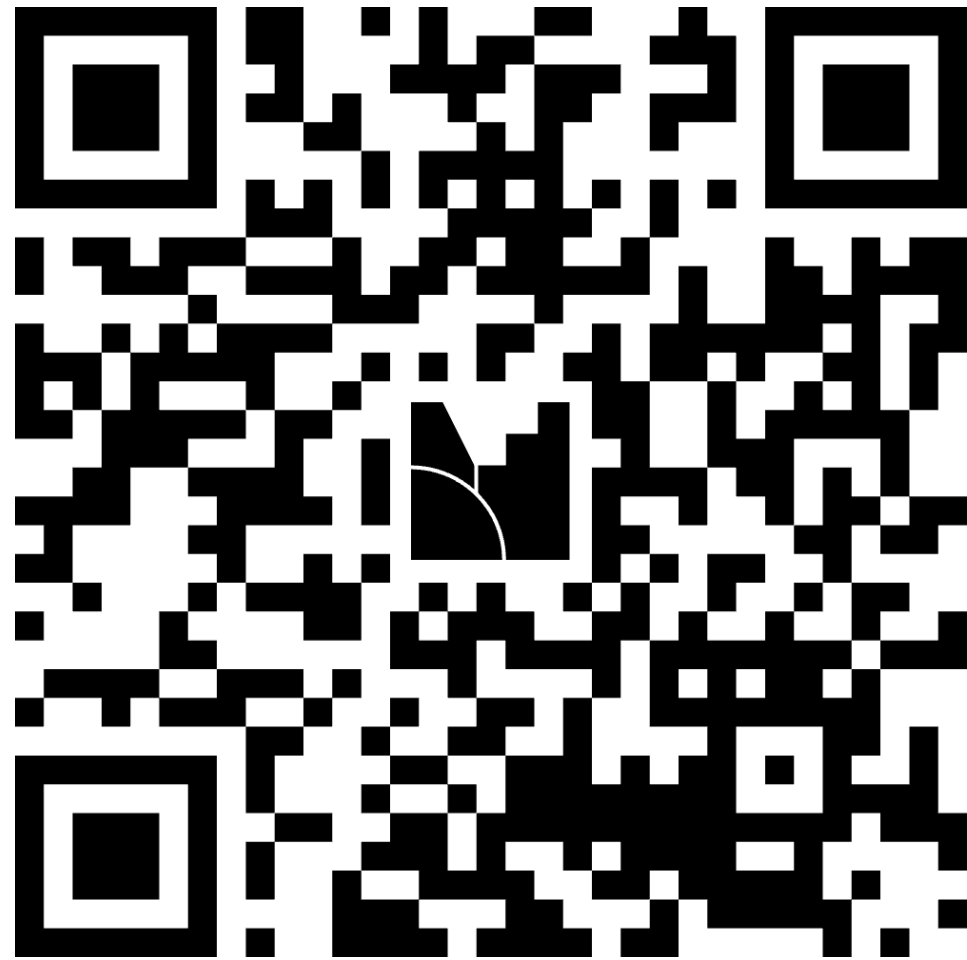
Gene expression

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Ligand 2	3.17	1.58	2.32	2.32	0.00	3.46
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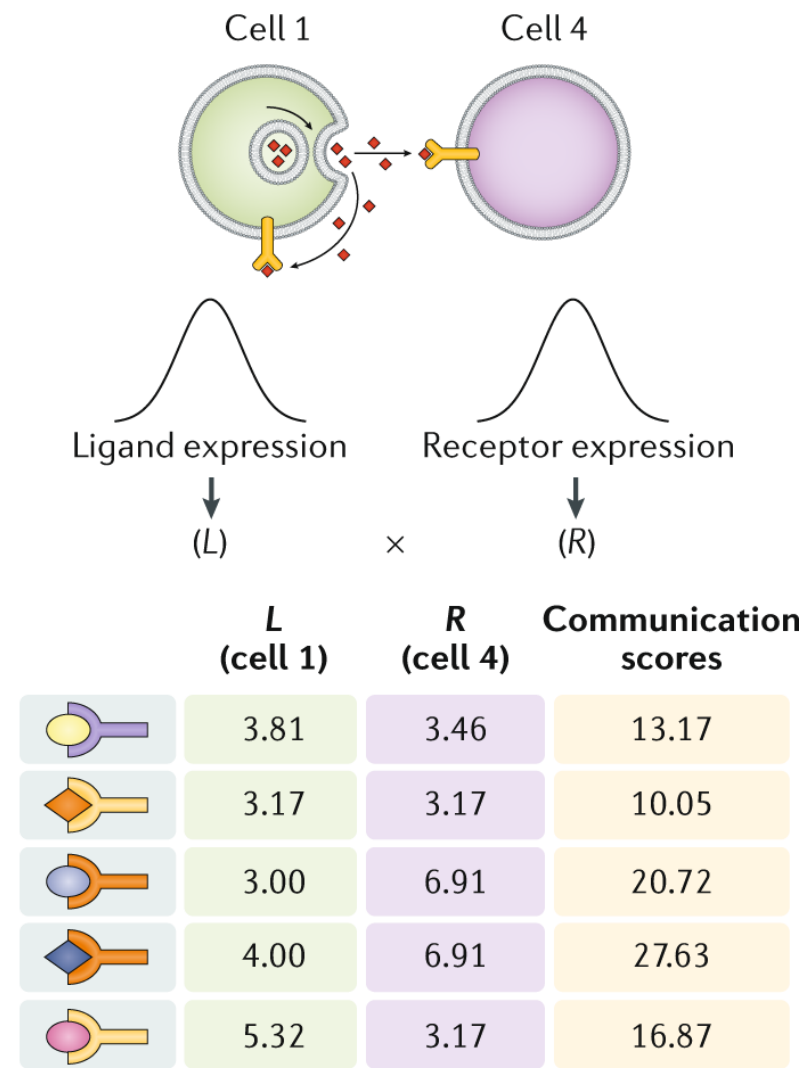
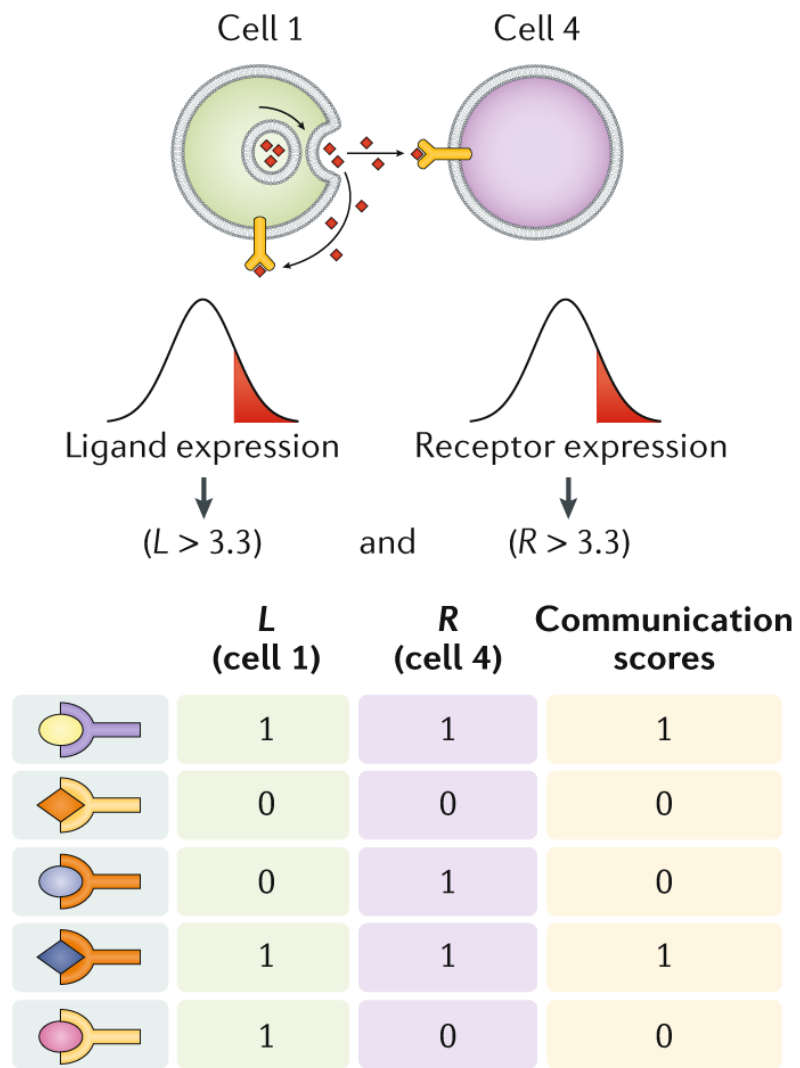
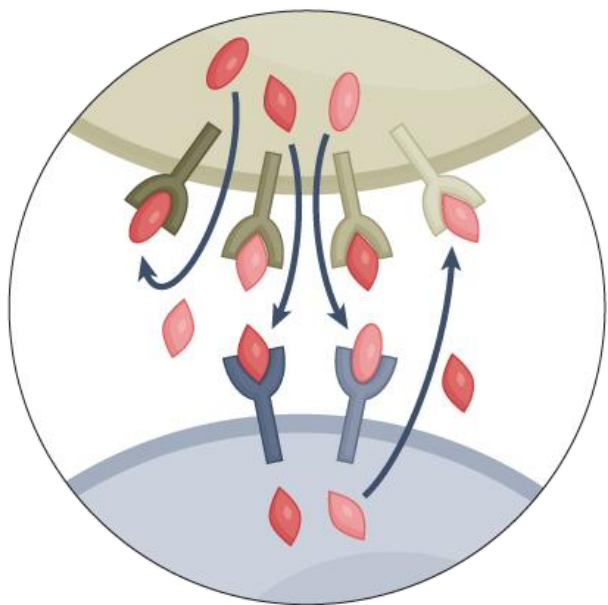
$F(\text{ligand}_i, \text{receptor}_j, \dots) \rightarrow \text{communication}_{i,j}$

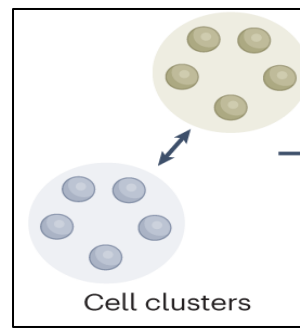
Receptor 1	4.70	4.58	4.17	3.46	3.81	3.58
Receptor 2	3.00	4.09	4.46	6.91	7.13	6.78
Receptor 3	4.32	2.00	3.00	3.17	1.00	3.81
	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6



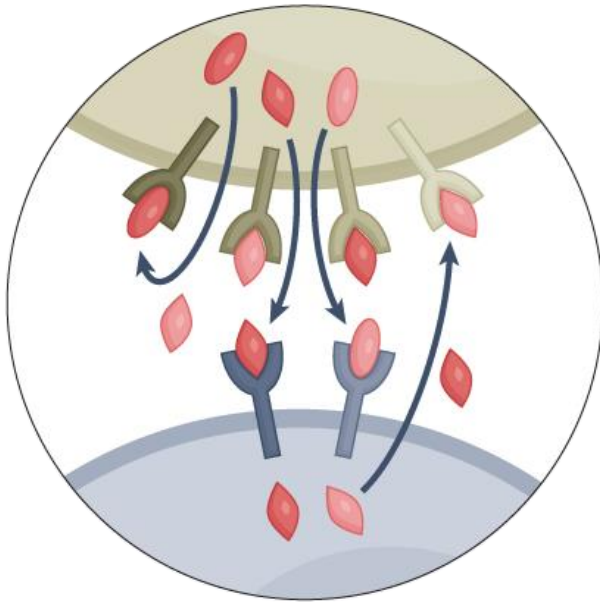


<https://www.menti.com/algauhs7u6f3>

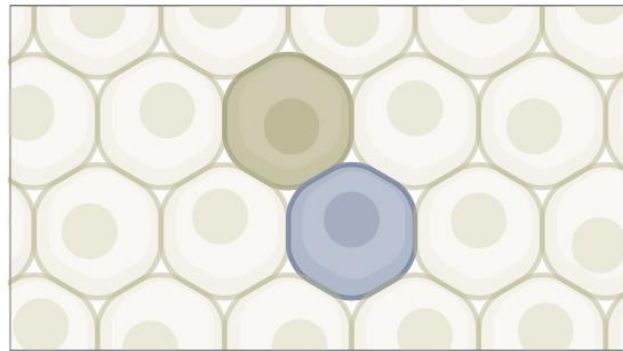




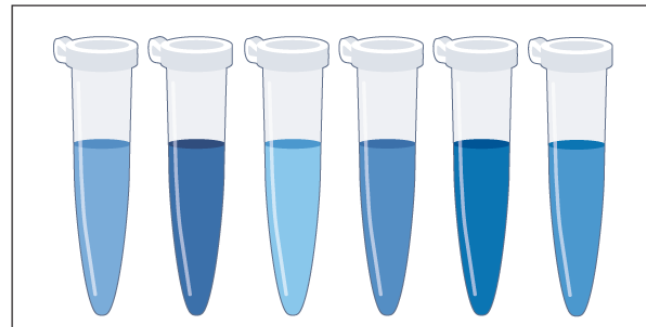
Cell type level



Spatially-aware



Spatially-informed



Differential CCC

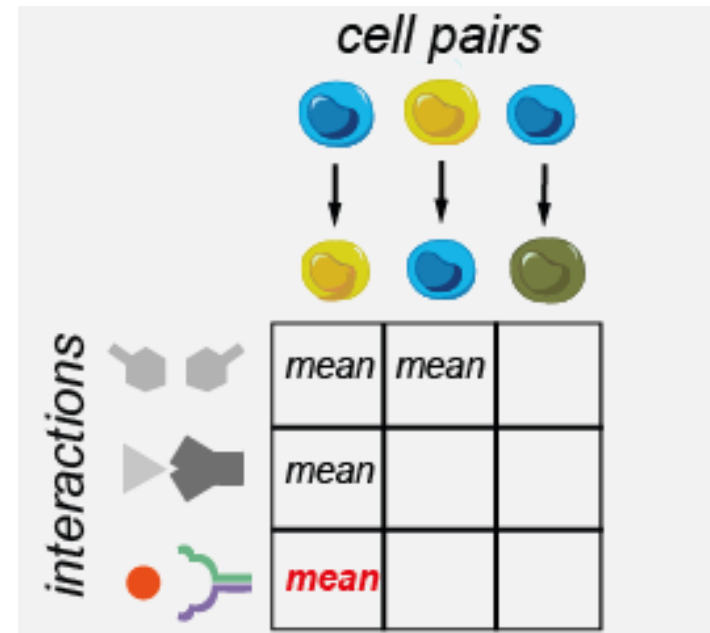
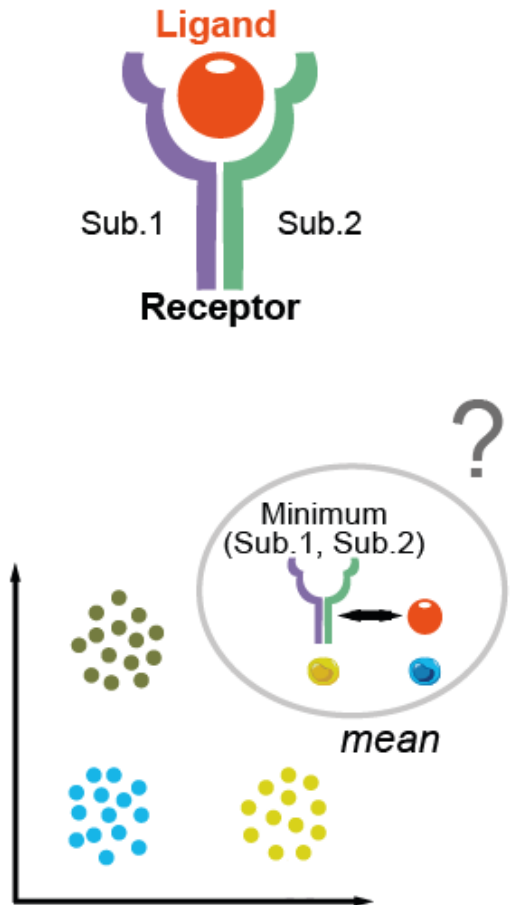
Cell type level: LIANA+

Method Name	Magnitude Score	Specificity Score
CellPhoneDB	lr_means	cellphone_pvals
Connectome	expr_prod	scaled_weight
log2FC	None	lr_logfc
NATMI	expr_prod	spec_weight
SingleCellSignalR	lrscore	None
Rank_Aggregate	magnitude_rank	specificity_rank
Geometric Mean	lr_gmeans	gmean_pvals
scSeqComm	inter_score	None
CellChat	lr_probs	cellchat_pvals

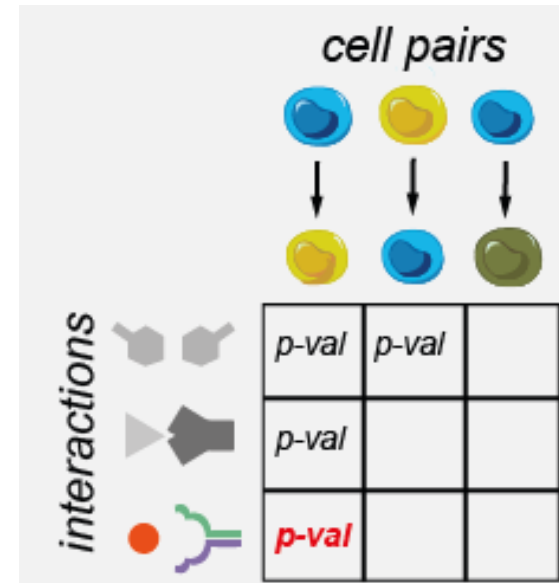
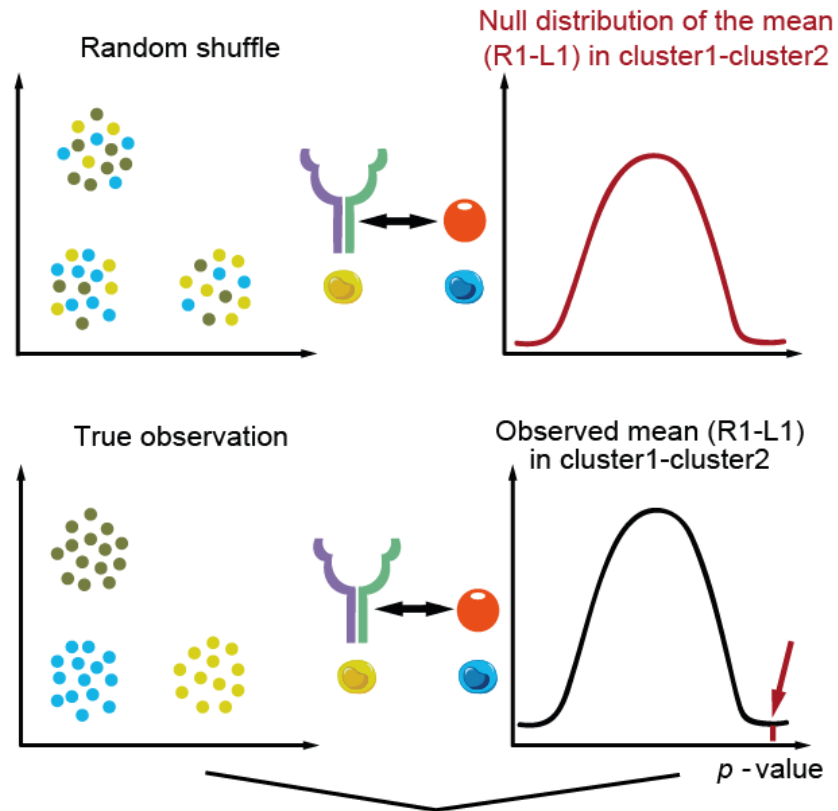
- The **magnitude score** quantifies the overall strength or likelihood of a ligand–receptor interaction between two cell types

- The **specificity score** reflects how uniquely an interaction is enriched in a given cell–cell pair compared to all others

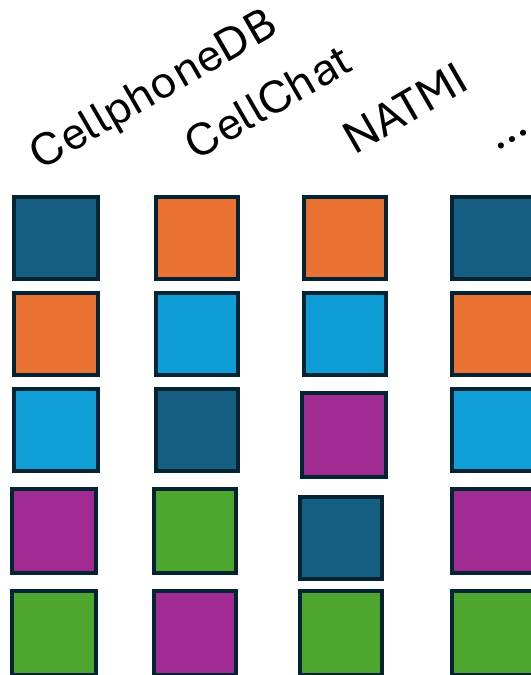
Cell type level: CellphoneDB



Cell type level: CellphoneDB

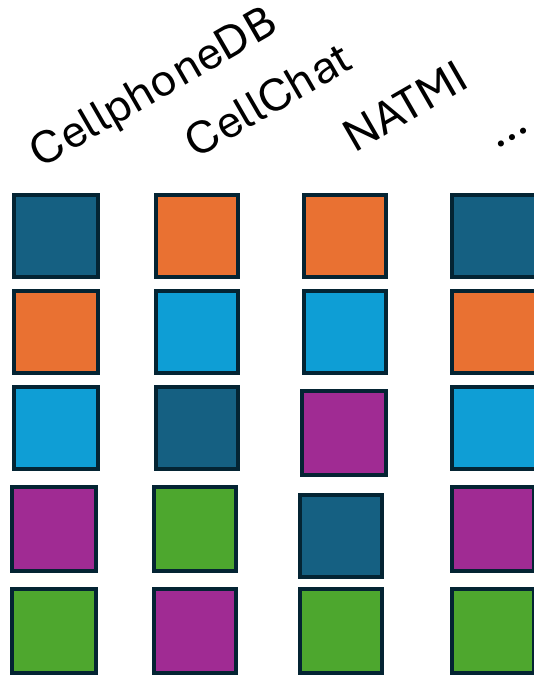


LIANA: Aggregating results



Most methods use a pair of scoring functions - where one often corresponds to the **magnitude (strength)** of interaction and the other reflects the **specificity** of a given interaction to a pair cell identities.

LIANA: Aggregating results



1. For each item i , collect its normalized ranks (0 = top, 1 = bottom):

→ $r_i = (r_{i1}, r_{i2}, \dots, r_{in})$

2. Sort Ranks

Sort each item's ranks from best to worst:

→ $r_{i(1)} \leq r_{i(2)} \leq \dots \leq r_{i(n)}$

($r_{i(1)}$ = best rank, $r_{i(k)}$ = k -th best)

3. Compute RRA Score (ρ_i)

For each k , calculate how unlikely it is to get rank $r_{i(k)}$ by chance:

→ $\beta_{k,n}(r_{i(k)})$ = Probability under random rankings (Beta distribution)

Then take the minimum across all k :

$\rho_i = \min_k \beta_{k,n}(r_{i(k)})$

→ Low ρ_i = Strong, consistent signal

→ High ρ_i = Luck or inconsistency

Spatially-aware

1.Spatially variable genes

2.Spatial proximity

Moran's I

$$I = \frac{N}{W} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

Nr of spots

Weight between spots i,j

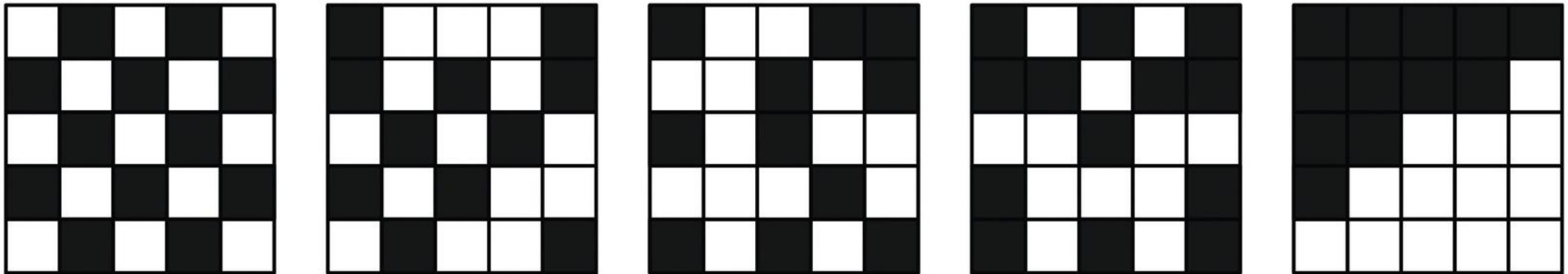
Deviation from the mean

Sum of weights

- If high expression occurs in contiguous clusters (high next to high, low next to low), **I > 0**.
- If high and low values alternate spatially, **I < 0**.
- If expression is random, **I ≈ 0**.

Moran's I

Moran's I



Moran's $I < E(I)$
indicates tend
to dispersion

Random
Moran's $I = E(I)$

Moran's $I > E(I)$
indicates tend
to clustering

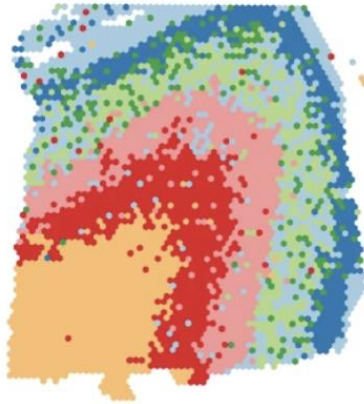
Spatially-aware

1.Spatially variable genes

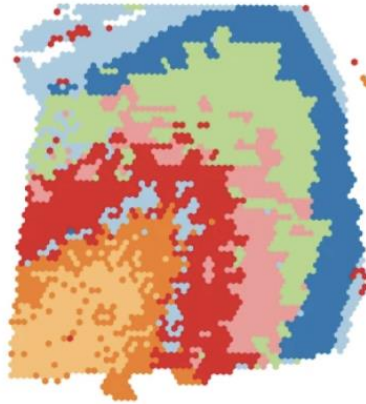
2.Spatial proximity

Restricting events to cells in close proximity (I)

Nonspatial, $\lambda = 0$ (0.382)



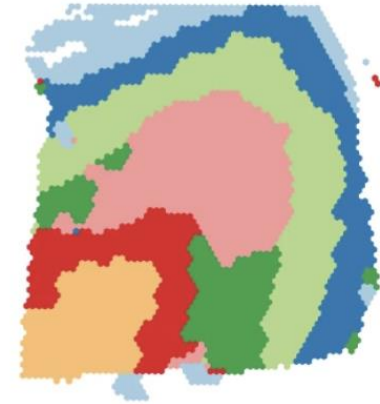
Giotto HMRF (0.340)



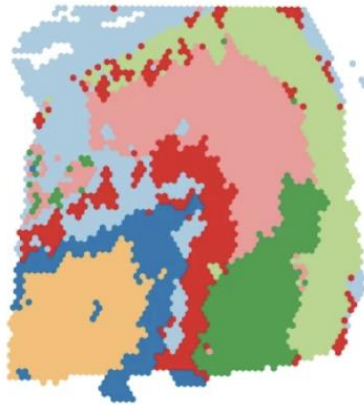
BayesSpace (0.540)



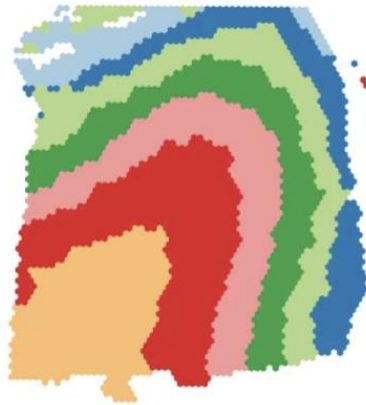
SpiceMix (0.433)



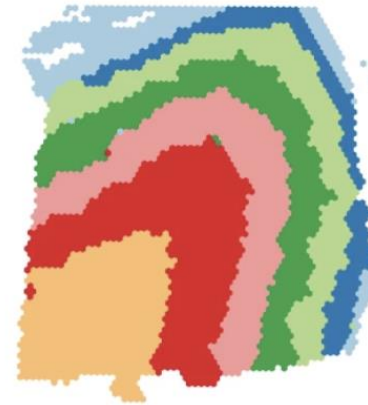
SpaGCN (0.289)



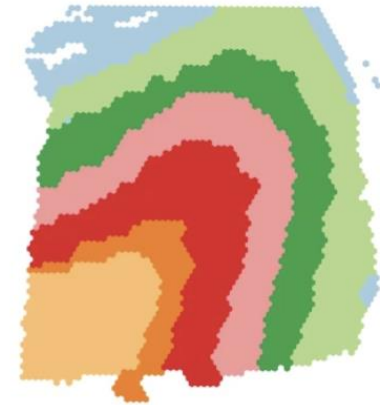
STAGATE (0.596)



GraphST (0.631)



BANKSY (0.566)

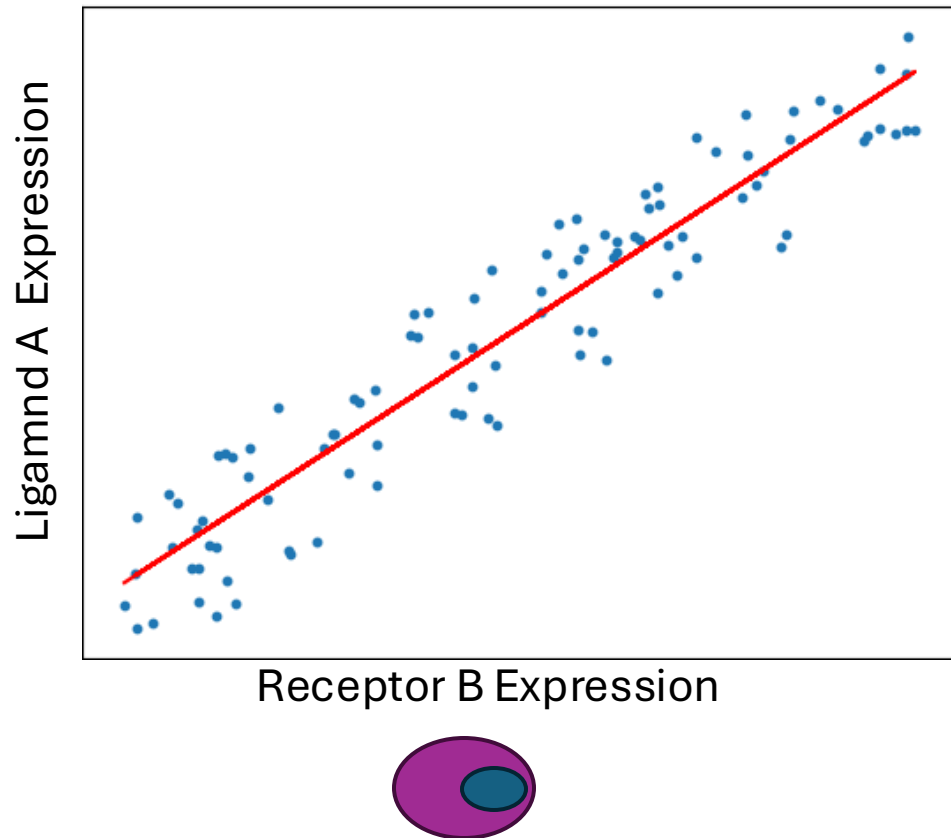


Restricting events to cells in close proximity (I)

```
In [ ]: sub_adata_domain = sub_adata[sub_adata.obs['spatial_domain_temp'] == domain]
sub_adata_domain
```

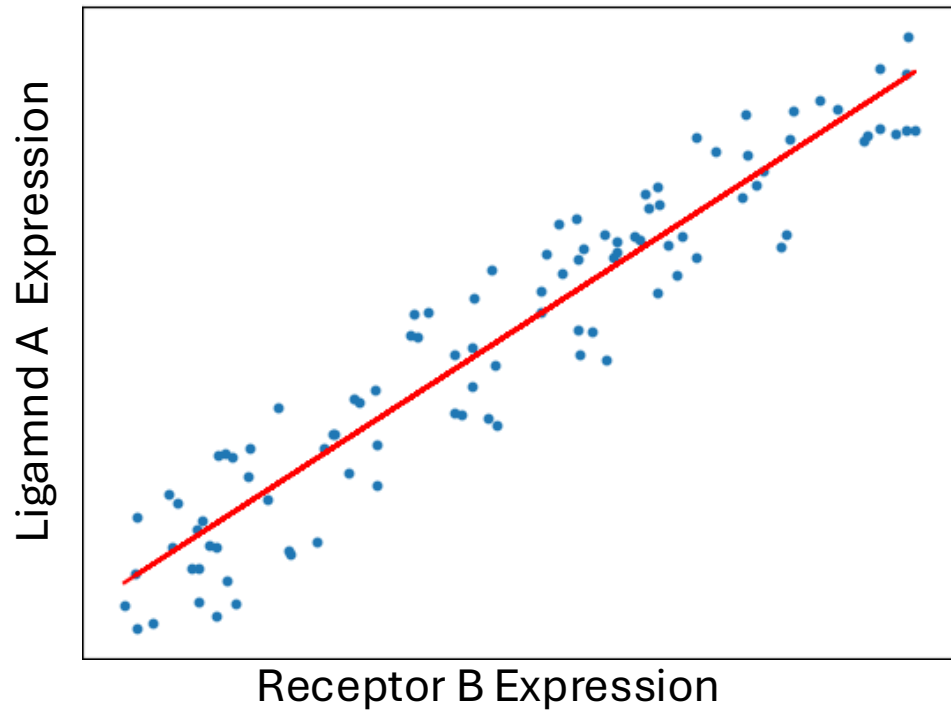
```
In [ ]: cellphonedb(sub_adata_domain,
                    groupby='cell_types',
                    # NOTE by default the resource uses HUMAN gene symbols
                    resource_name='mouseconsensus',
                    expr_prop=0.1,
                    verbose=True,
                    use_raw = False,
                    layer = 'counts',
                    key_added='cpdb_res')
```

Restricting events to cells in close proximity (II)

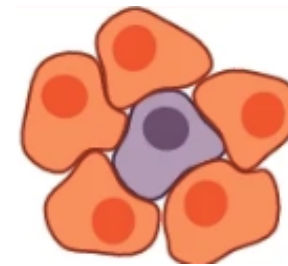


Are these two cells communicating through A-B signalling?

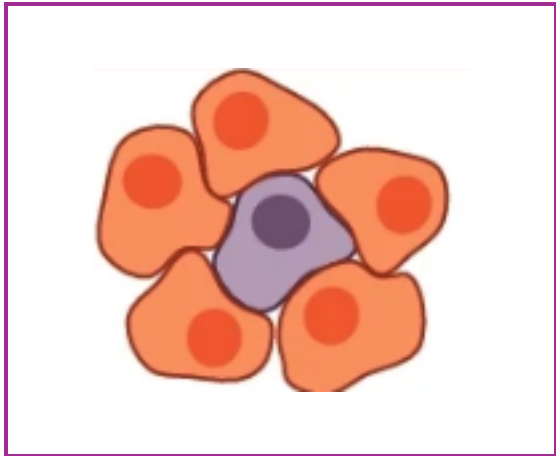
Restricting events to cells in close proximity (II)



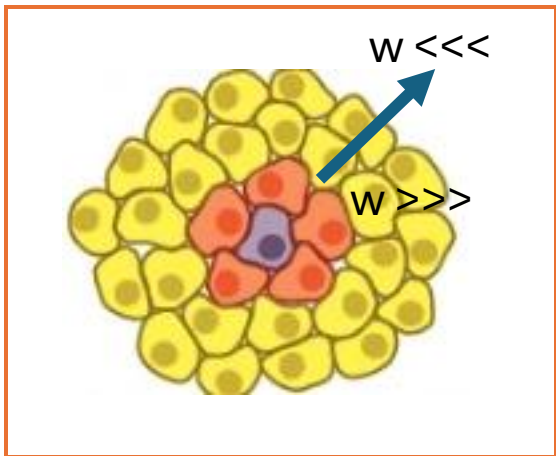
Are these two cells communicating through A-B signalling? **Maybe**



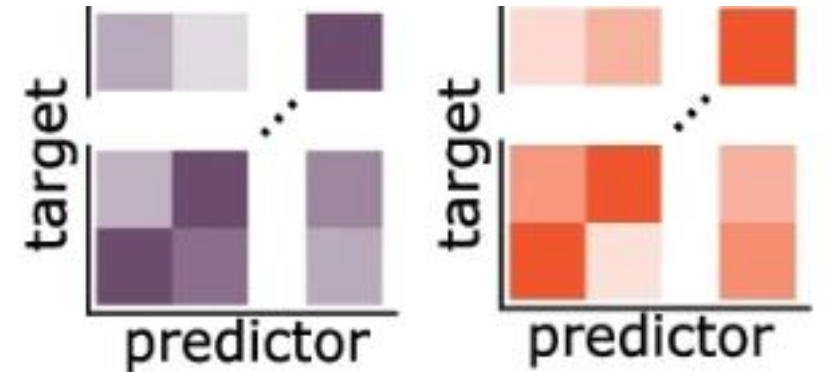
Restricting events to cells in close proximity (II)



Receptor $\sim$ linear_model.fit(ligand neighbors)

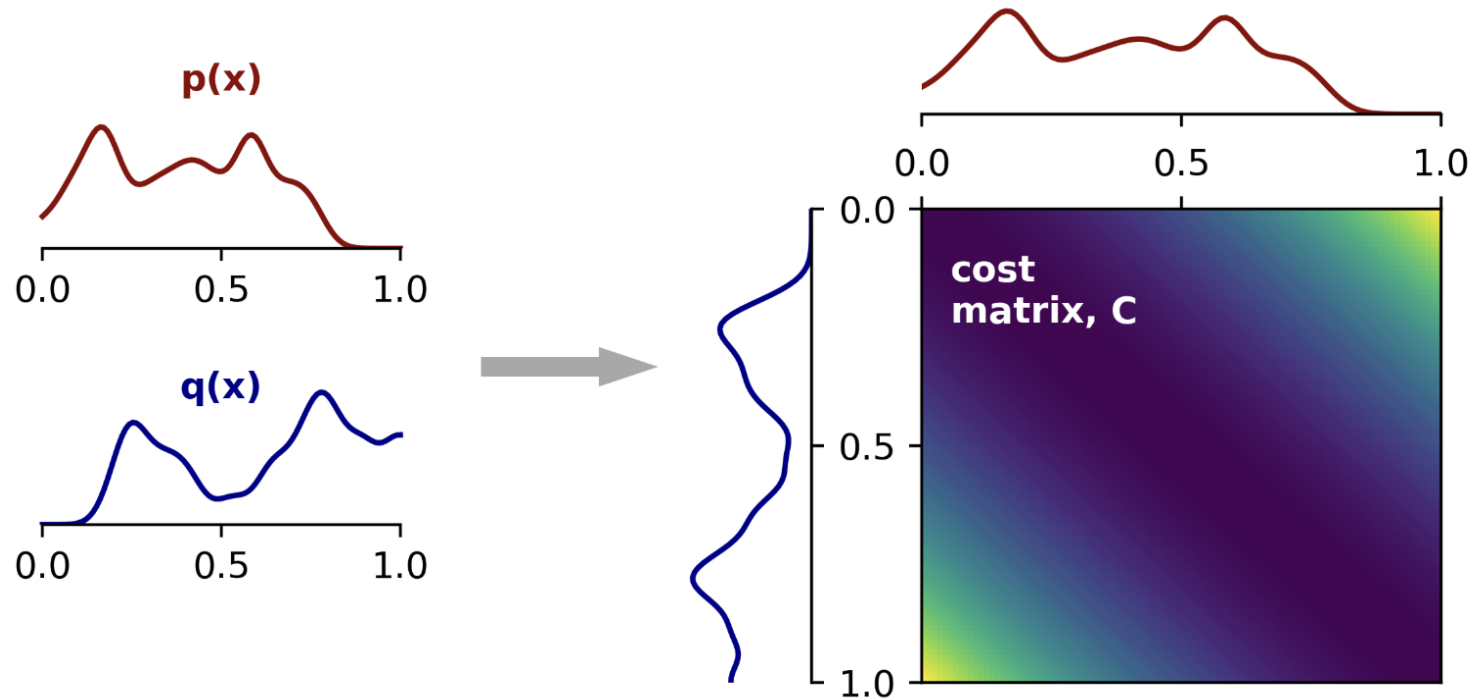


Receptor $\sim$ linear_model.fit($w * \text{ligands}$)



Spatially-informed

Optimal Transport primer

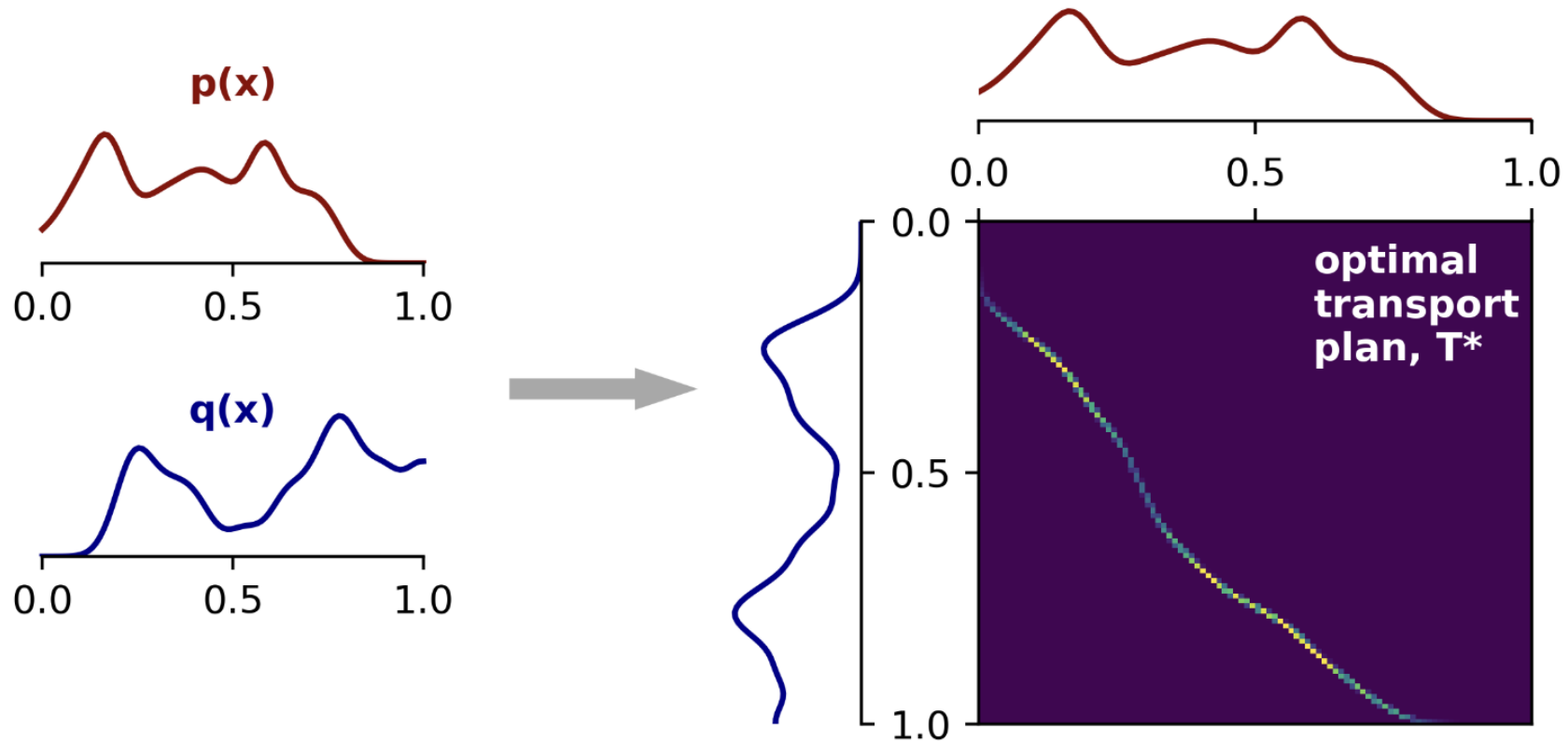


We find a *coupling* P_{kl} :

$$\text{minimize } \sum_{k,l} C_{kl} P_{kl} \quad \text{such that} \quad \sum_l P_{kl} = a_k, \sum_k P_{kl} = b_l.$$

P_{kl} = strength of potential communication between cells k and l .

Optimal Transport primer



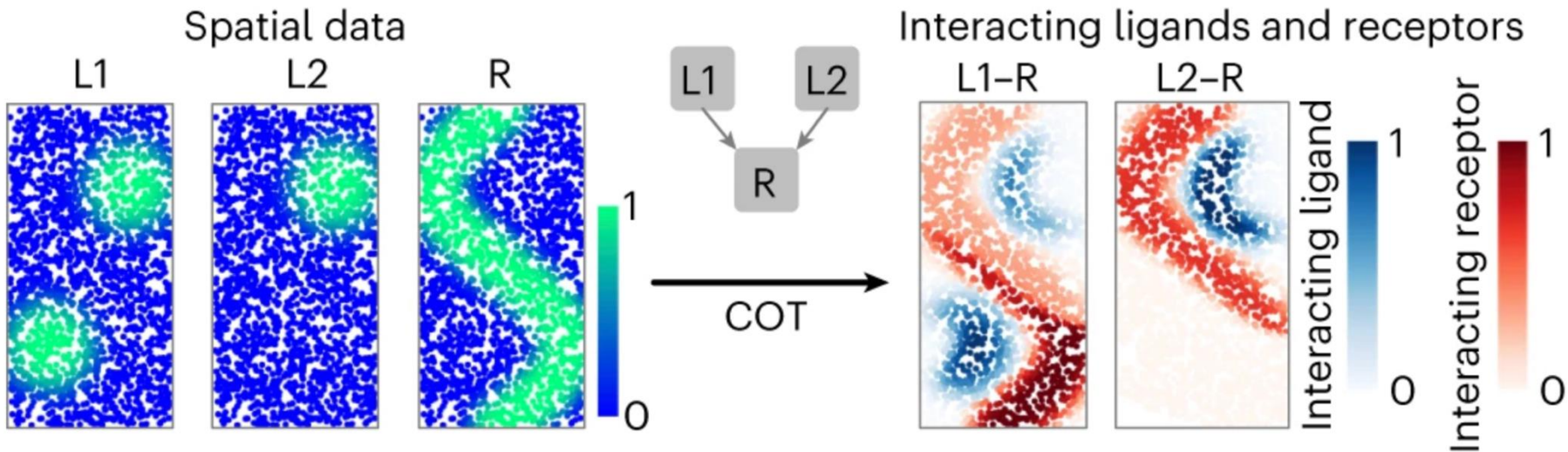
COMMOT

Classical OT:

- Normalization problem
- No competition or multi-pairing
- No multi-species coupling
- Spatial cutoff missing

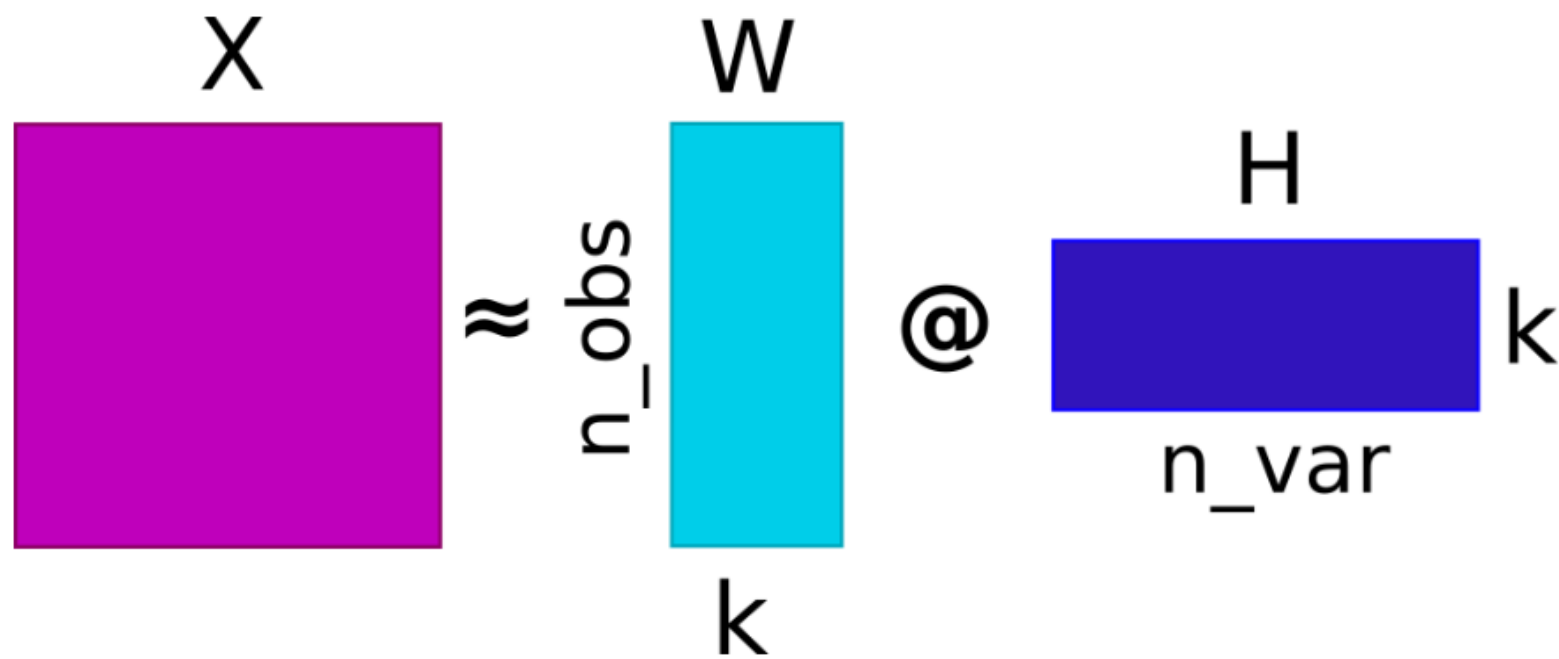
COMMOT:

- Real expression magnitudes
- Include multiple pairs at once, so the same receptor mass can't be reused across pairs
- Add spatial range limits

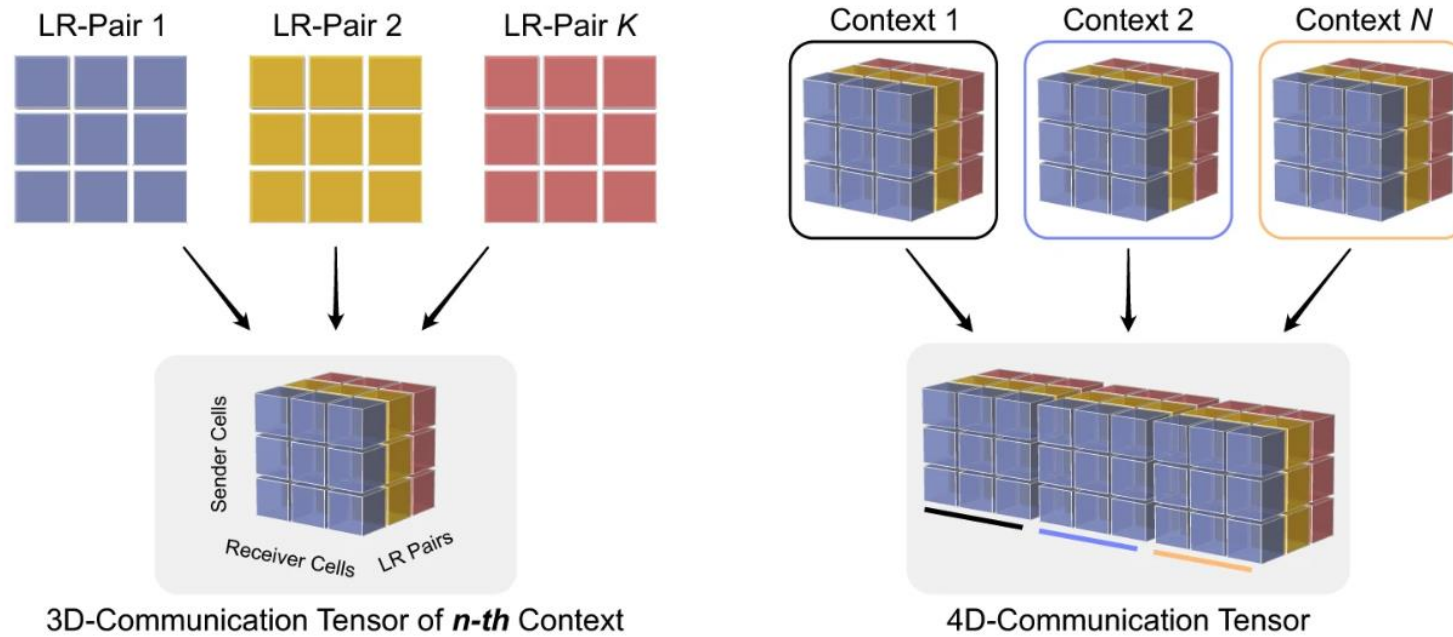


Differential CCC

NMF primer

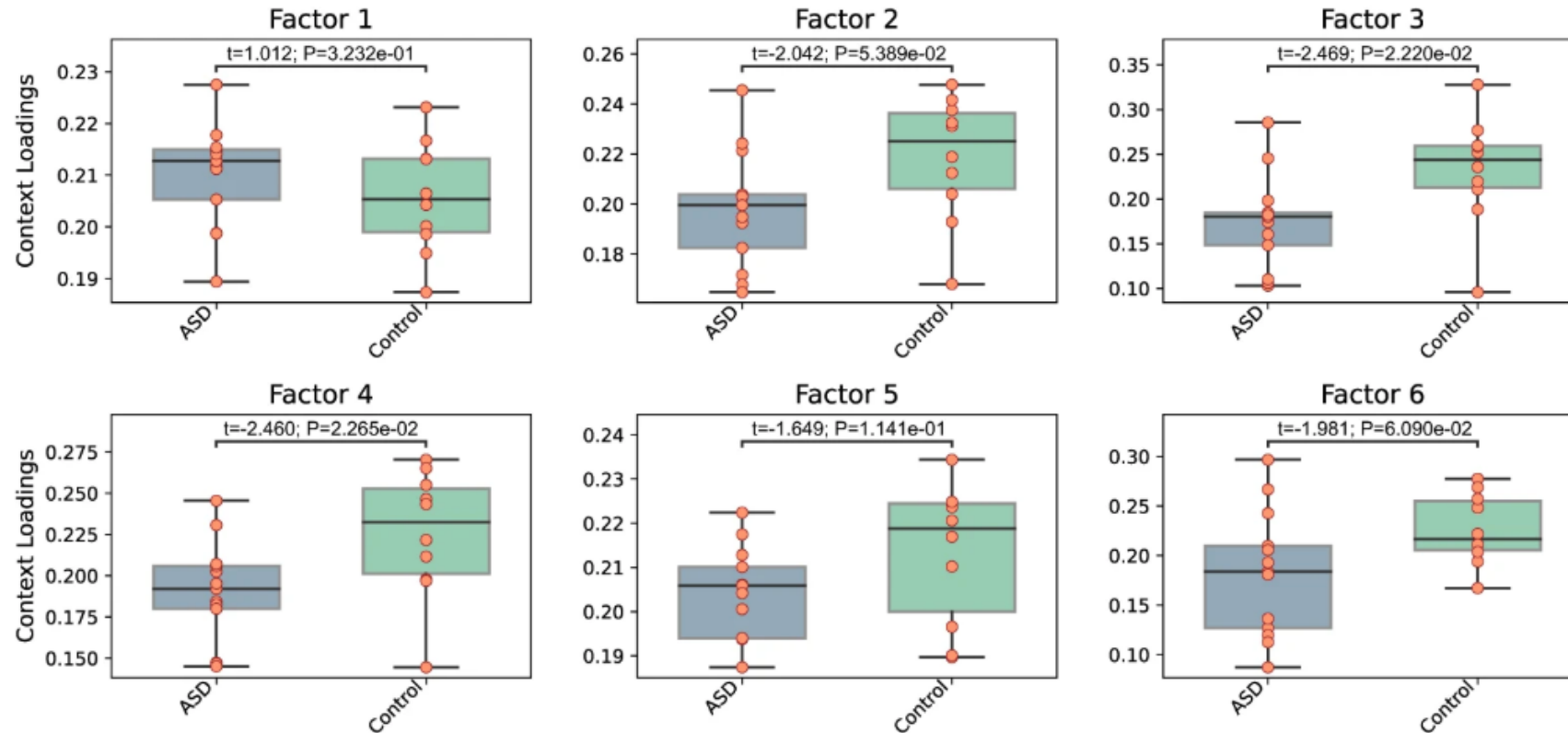


Tensor factorisation

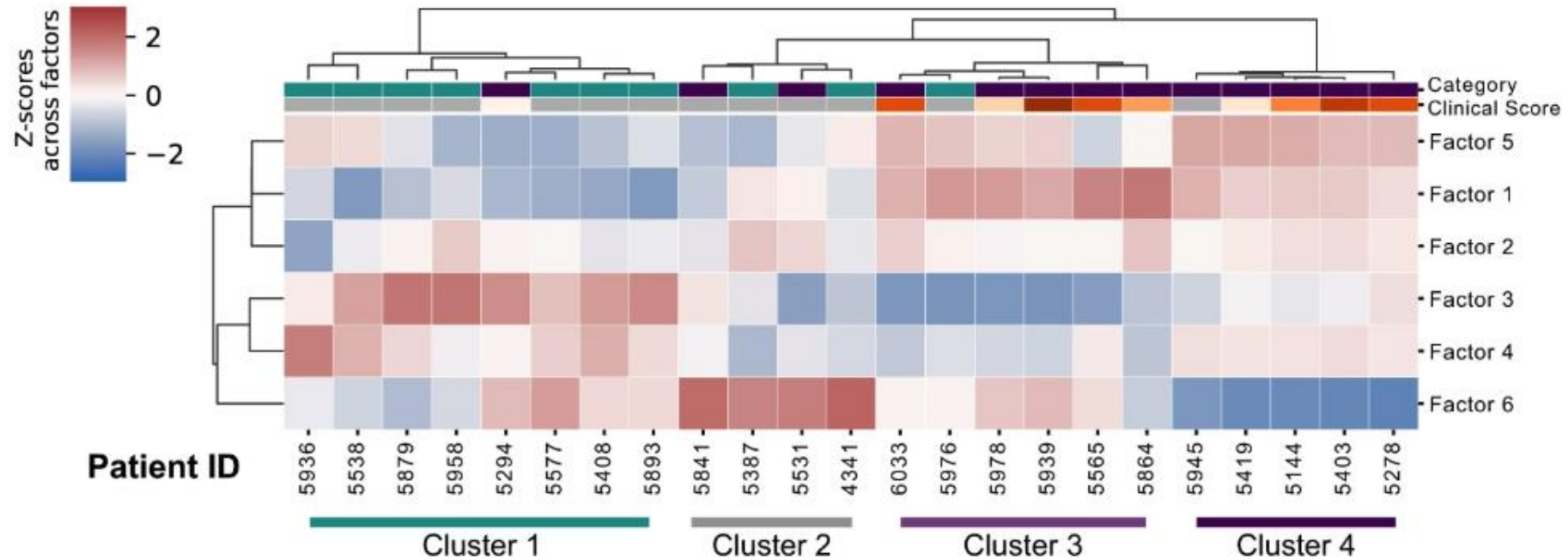


$$\text{Factor 1} + \text{Factor 2} + \dots + \text{Factor } R$$

Downstream analyses



Downstream analyses



ADPKD application

Prevalence 1:1,000-3,000

Fluid-filled cysts in kidneys

Ultrasound - 20-30 years

Renal failure - 55-65 years

Extra-renal manifestations

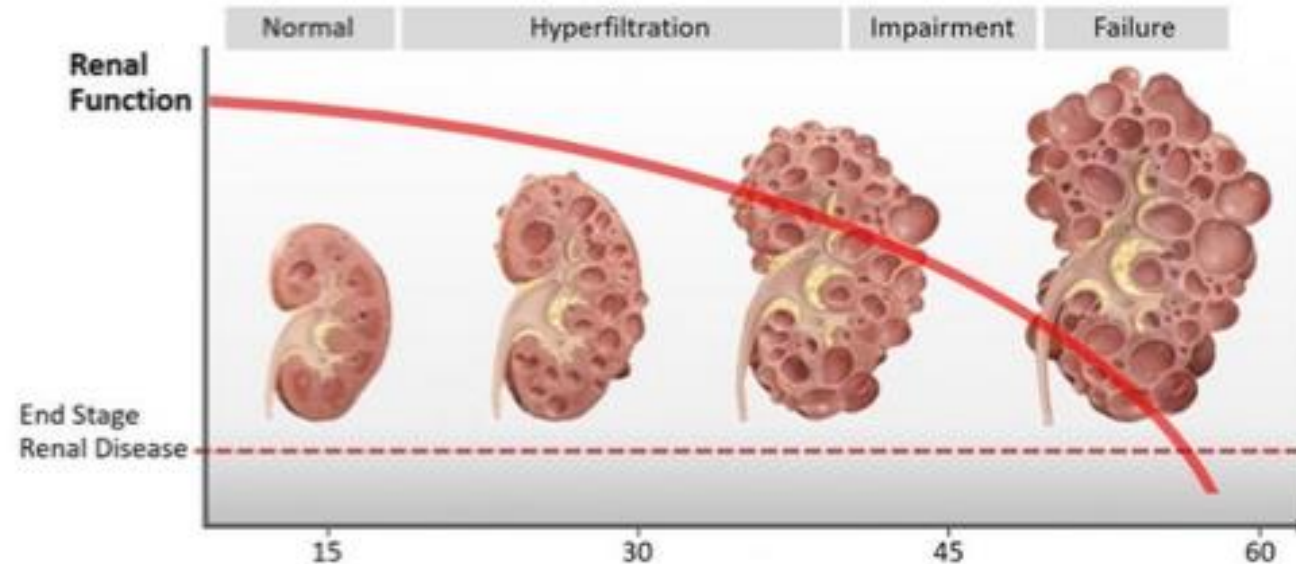
Hypertension

Cerebral aneurysms

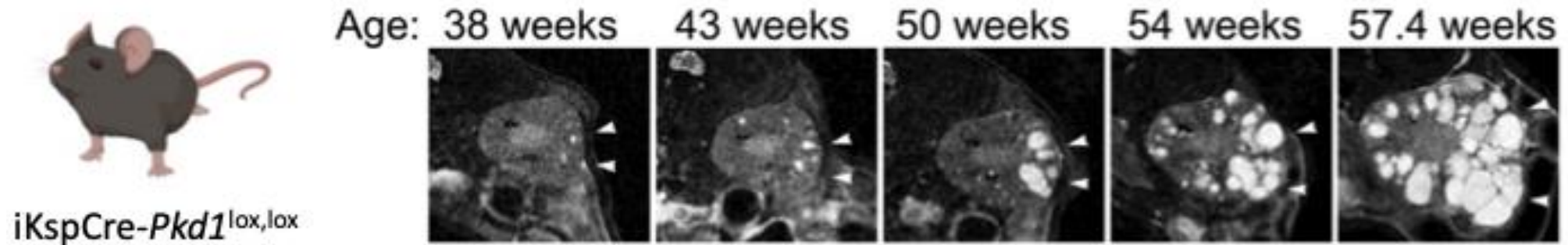
Cardiovalvular abnormalities

Cysts in liver, pancreas

Disease Progression in ADPKD



ADPKD application



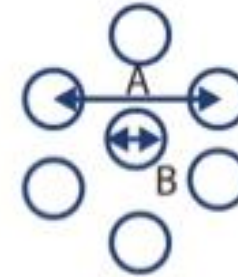
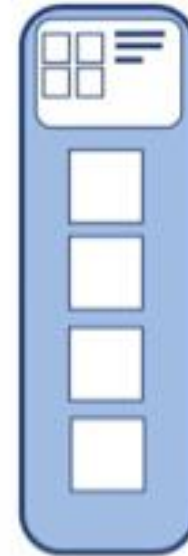
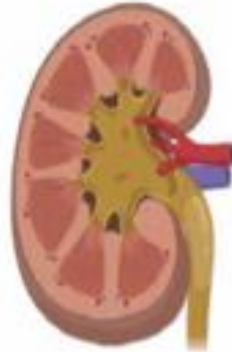
Treating the mice with a low dose of tamoxifen leads to a slow-progressing cystic disease, which resembles human ADPKD.

ADPKD application

iKsp-*Pkd1*^{del}
no tamoxifen

264-297 days

iKsp-*Pkd1*^{del}
12.5 mg/kg tamoxifen



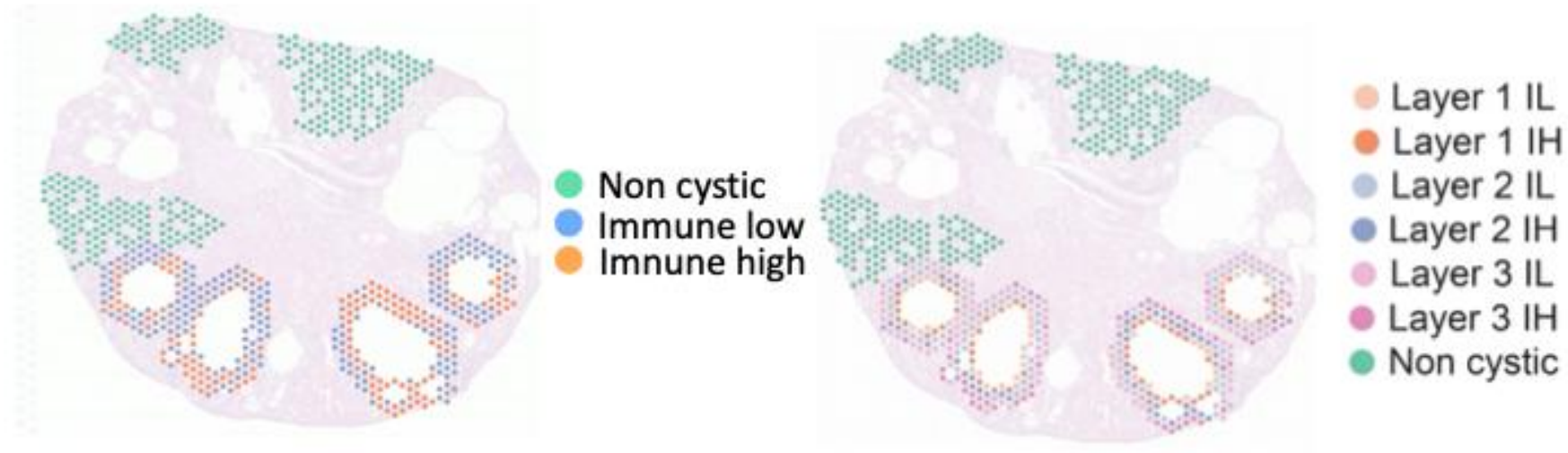
A: 100 μm

B: 55 μm

ADPKD application

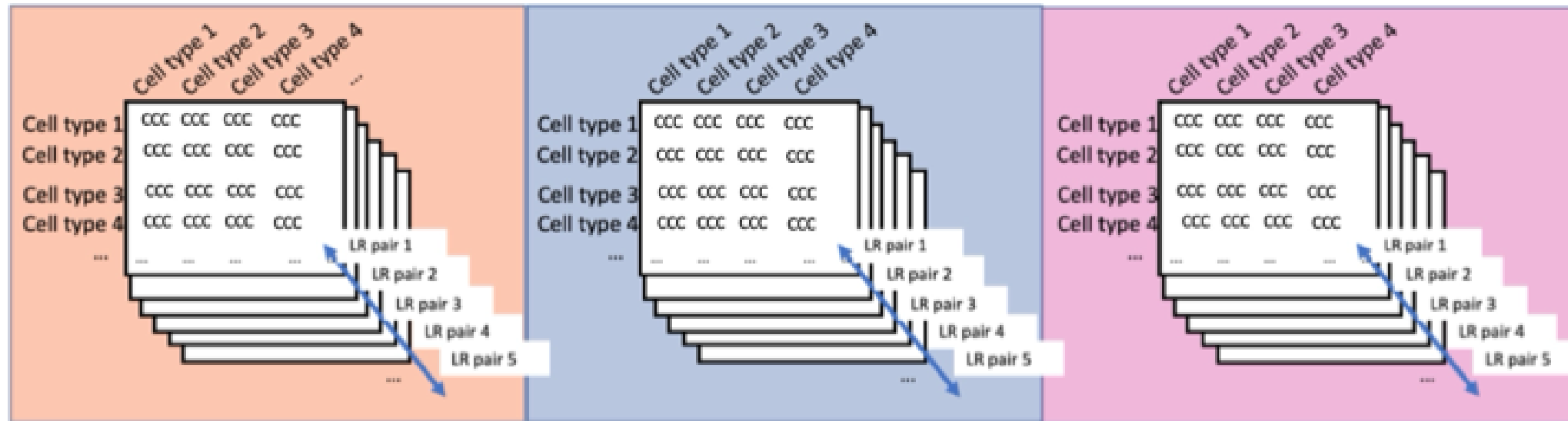
How do cyst lining cells communicate with neighboring cells? What cell types are involved?

ADPKD application



We computed cell-cell communication scores (CCI) between pairs of unsupervised clusters in the **Immune Low (IL) context** using LIANA+

ADPKD application

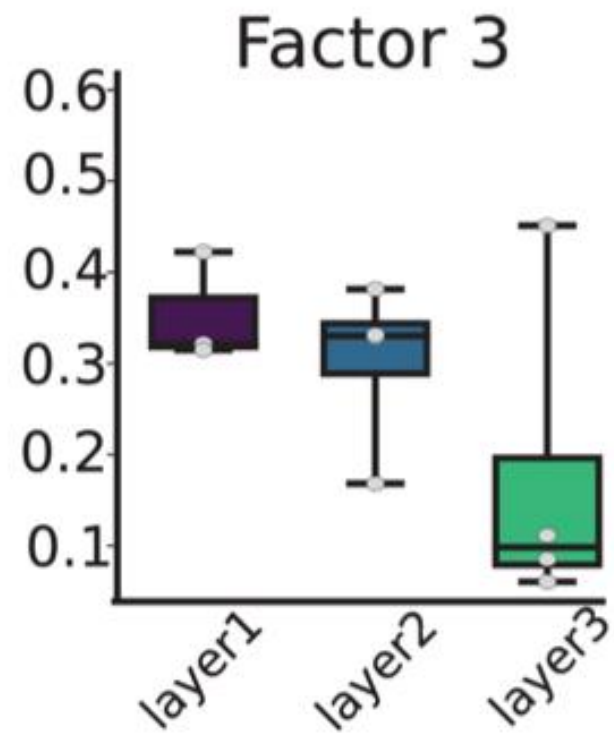
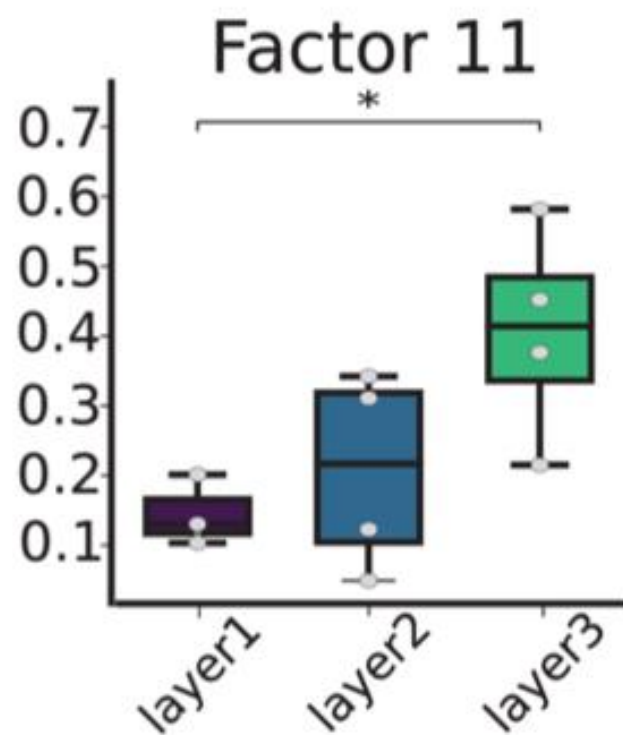


We computed cell-cell communication scores (CCI) between pairs of unsupervised clusters in the **Immune Low (IL) context** using LIANA+

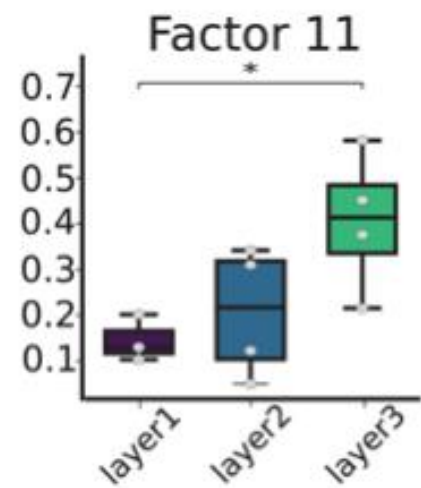
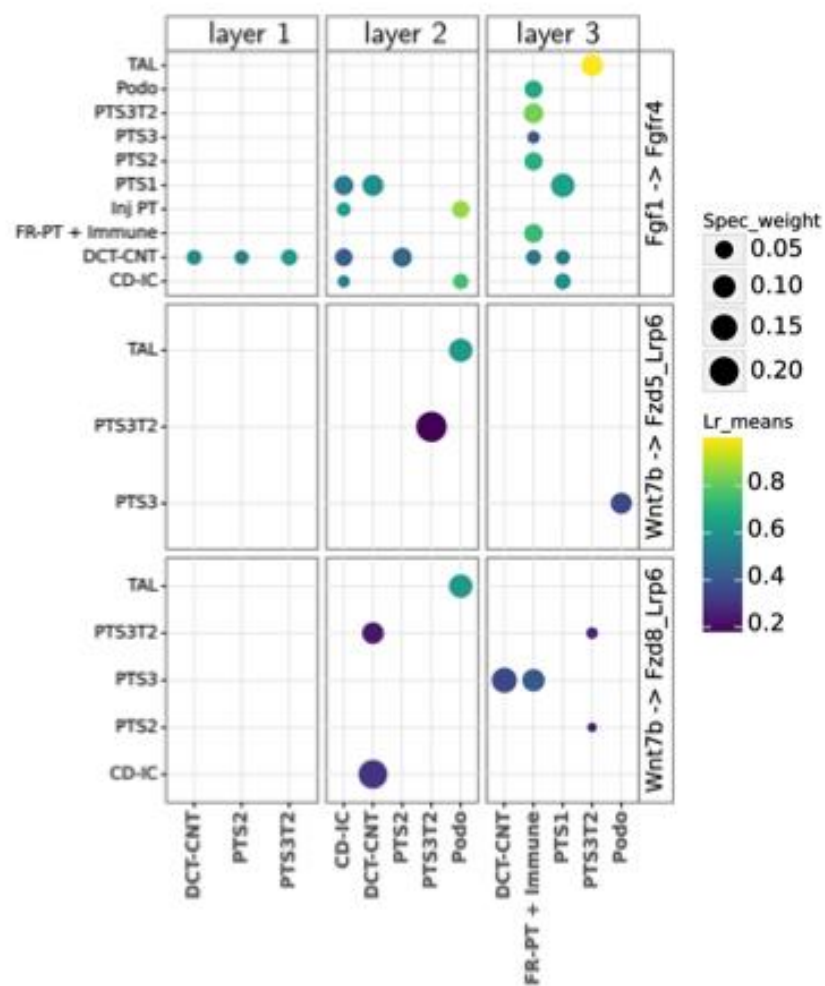
ADPKD application

	L-R pairs	Layer	Sender cells	Receiver cells
Factors				

ADPKD application

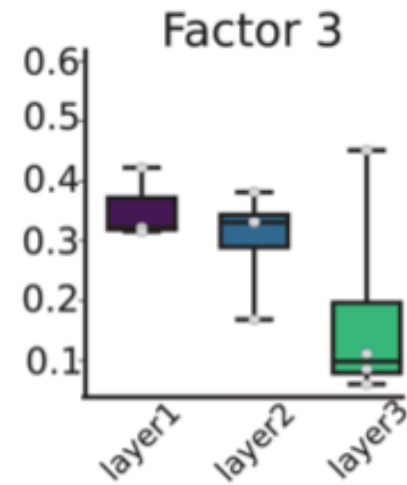
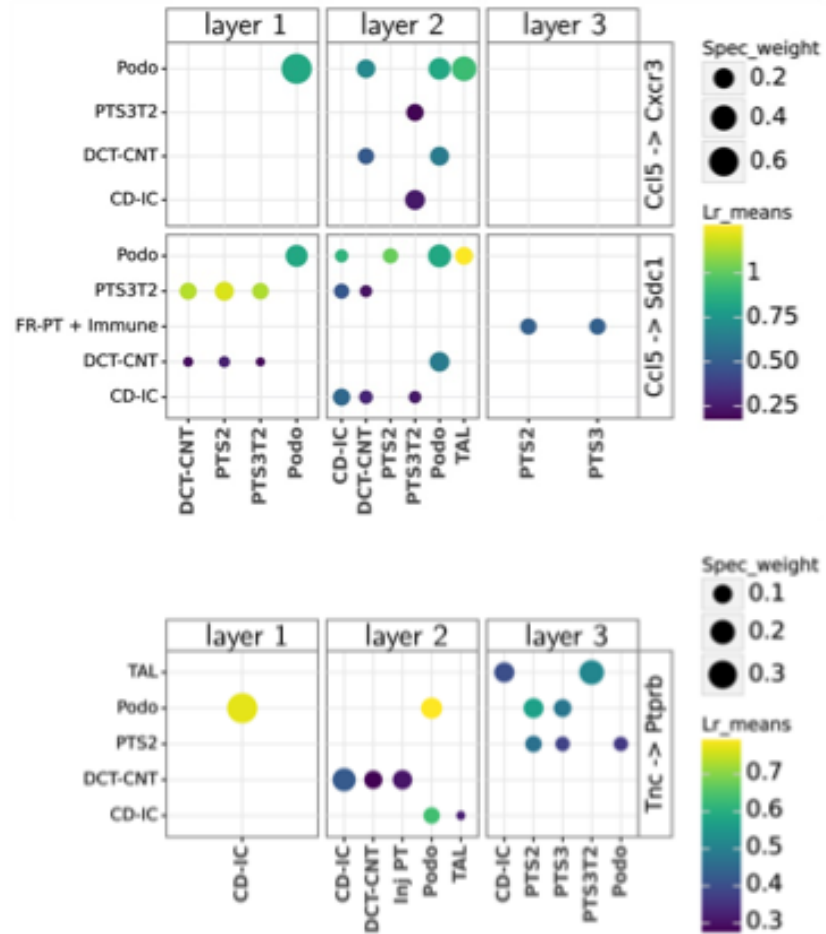


ADPKD application



Reduced Wnt and Fgf signaling

ADPKD application



ECM remodeling and immune recruitment

Summary

- **Single-cell CCC methods** can often be **adapted** to include spatial context (e.g., SVGs, Domains, ROIs...)
- Some tools are **built for spatial data from the ground up** (e.g., **COMMOT**, **SpaOTsc**, **Giotto**, etc.).
- **Validation remains challenging:**
 - No comprehensive *ground truth* datasets exist.
 - Current results should be viewed as **hypothesis-generating**, not definitive.
- Remember: CCC inference is (usually) based on **transcript expression**, not direct measurement of signaling or protein activity.
- Cell–cell communication inference is still largely a guessing game with many **false positives**. Adding spatial information increases confidence in our predictions